

AVR

AVR

- Automatic Voltage Regulator
- A device or equipment that monitor power supply.
- It will automatically correct voltage fluctuation to ensure reliable and steady supply of required power
- protection against over voltage, over current
- 0.16 seconds response time upon detection of power problem



UPS

UPS

- Uninterruptible Power Supply
- Provides continuous power supply in the event of power interruption
- Improves and regulates fluctuations to prevent damage to sensitive loads
- a power supply that includes a battery to maintain power in the event of a power outage.
- Typically, a UPS keeps a computer running for several minutes after a power outage, enabling you to save data and shut down the computer gracefully



Some Symptoms of Power Problem

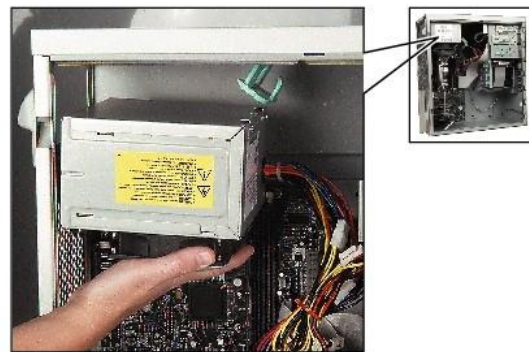
- The computer “freezes up” with the cause not being traced
- Random memory errors
- Lost data on the hard disk
- Damage chips on the circuit board

Steps in Power Protection

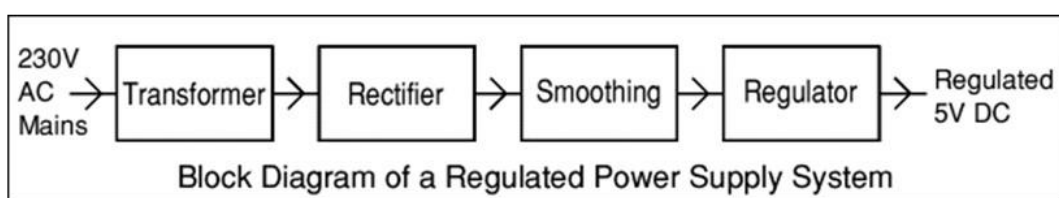
- Check that your outlets are wired correctly
- Check what else connected into your line
- Remove large motors, air conditioners, refrigerators, heater and power tools near the PC
- Ensure common ground among devices

Power Supply

- is a reference to a source of **electrical power**
- Converts (220/110) into various voltage readings required by the computer
- A device or system that supplies electrical or other types of energy to an output load or group of loads is called a **power supply unit** or **PSU**
- computer power supply typically is designed to convert 110-240 V AC power from the mains, to several low-voltage DC power outputs for the internal components of the computer



Power Supply Block Diagram



For example a 5V regulated supply:

Transformer - steps down high voltage AC mains to low voltage AC

Rectifier - converts AC to DC, but the DC output is varying.

Smoothing - smooths the DC from varying greatly to a small ripple.

Regulator - eliminates ripple by setting DC output to a fixed voltage

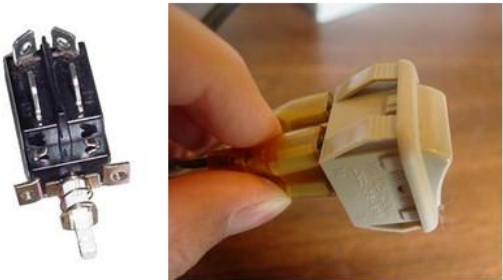
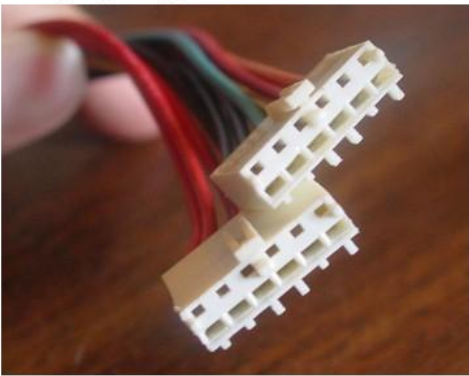
Power Supplies: How Much Power Do You Need?

- For overall power supply wattage, add the requirement for each device in your system, then multiply by 1.5 - Furthermore, power supplies are more efficient and reliable when loaded to 30% - 70% of maximum capacity	Component	Requirement
	AGP Video Card	30W - 50W
	PCI Express Video	100W - 225W
	Average PCI Card	5W - 10W
	DVD/CD	20W - 30W
	Hard Drive	15W - 30W
	Case/CPU Fans	3W (ea.)
	Motherboard (w/o CPU or RAM)	50W - 100W
	RAM	15W per 1GB
	Pentium III Processor	40W
	Pentium 4 Processor	80W - 125W
	AMD Athlon Processor	80W - 125W

Types of Power Supply

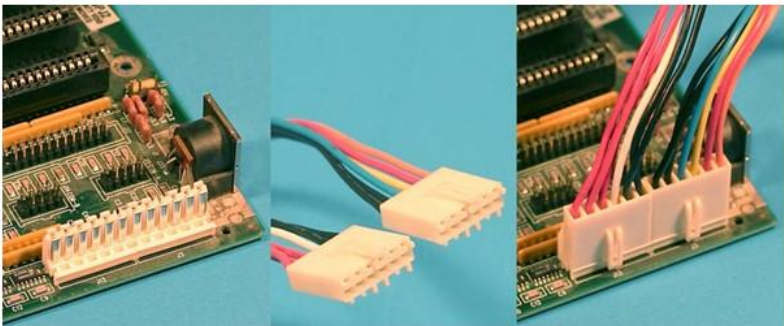
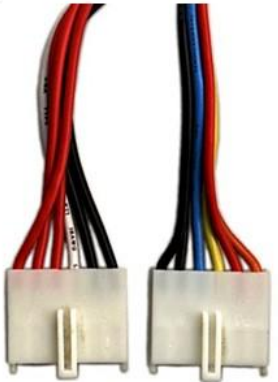
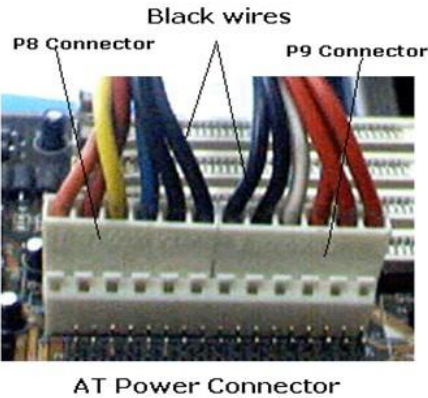
AT (Advance Technology) Power Supply

- These are the older of the power supplies. They have two 6-pin keyed connectors that plug into the motherboard right next to each other
- use a normal on-off switch, which directly turns the power supply on or off
- Power-on switch wire from the front of the computer is connected directly to the power supply



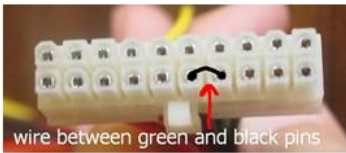
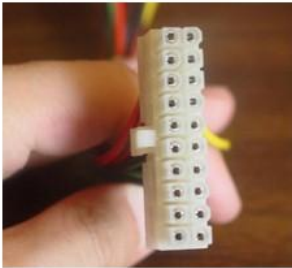
AT power connector

Color	Pin	Signal
Orange	P8.1	Power Good
Red	P8.2	+5 V
Yellow	P8.3	+12 V
Blue	P8.4	-12 V
Black	P8.5	Ground
Black	P8.6	Ground
Black	P9.1	Ground
Black	P9.2	Ground
White	P9.3	-5 V
Red	P9.4	+5 V
Red	P9.5	+5 V
Red	P9.6	+5 V

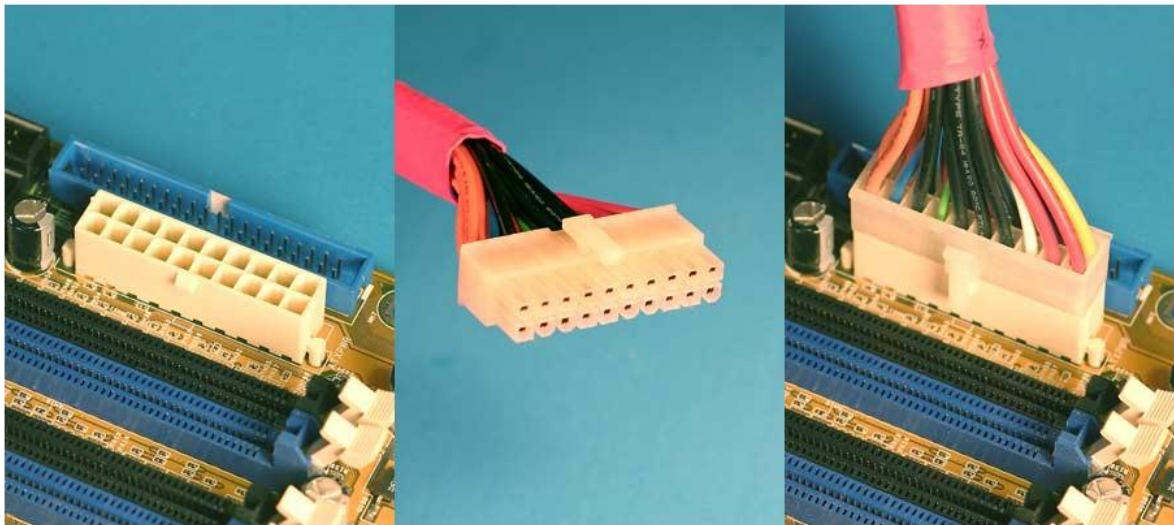


"Classic" ATX Power Supply

- Advanced Technology Extended
- was created by Intel in 1995
- had a power button that was directly connected to the system computer power supply
- is a single 20-pin connector that only plugs in one way
- An ATX power supply does not directly connect to the system power button, allowing the computer to be turned off via [software](#).
- However, many ATX power supplies have a manual switch on the back to ensure the computer is truly off and no power is being sent to the components



ATX 20-pin Power Cable Connector



ATX 20-pin Power Cable Connector

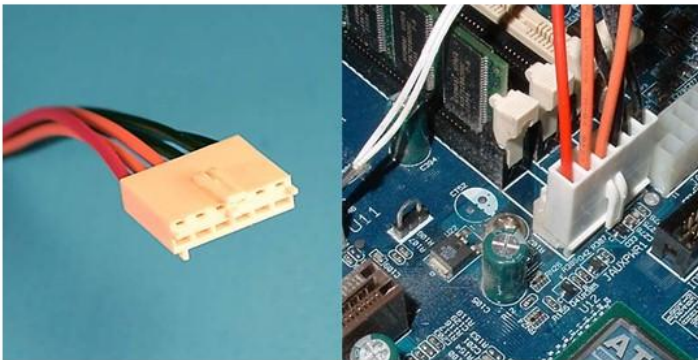
20 Pin ATX Pinout					
Pins 1 through 10			Pins 11 through 20		
Description	Wire color	Pin number	Pin number	Wire color	Description
+3.3 volts	orange	1	11	orange	+3.3 volts
+3.3 volts	orange	2	12	blue	-12 volts
ground	black	3	13	black	ground
+5 volts	red	4	14	green	PS_ON#
ground	black	5	15	black	ground
+5 volts	red	6	16	black	ground
ground	black	7	17	black	ground
PWR_OK	gray	8	18	white	-5 volts (optional)
VS _B +5 volts	purple	9	19	red	+5 volts
+12 volts	yellow	10	20	red	+5 volts

ATX Version

- **ATX Version 2.01** - 20 pin block power connector
- **ATX Version 2.03** – 20 pin block power connector, P6 and +12V 4-pin Auxiliary connector

ATX Version 2.03

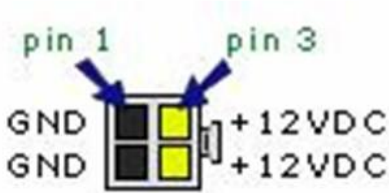
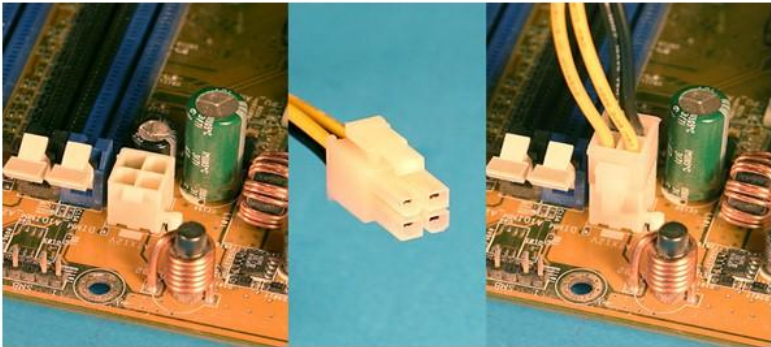
- 6 pin auxiliary power cable
- The aux power cable was added to provide extra wattage to motherboards
- It's most commonly found on older dual CPU AMD motherboards.



Pinout		
Pin number	Wire color	Description
1	black	ground
2	black	ground
3	black	ground
4	orange	+3.3 volts
5	orange	+3.3 volts
6	red	+5 volts

ATX Version 2.03

- Auxiliary connector was designed to deliver up to 58 watts of additional +3.3V and +5V power to power-hungry motherboards, which often needed the additional power for CPU, memory and AGP slot voltage regulators.
- 4 pin ATX +12 volt power cable
- If your power supply doesn't have a 4 pin 12 V cable then you can provide one with the adapter



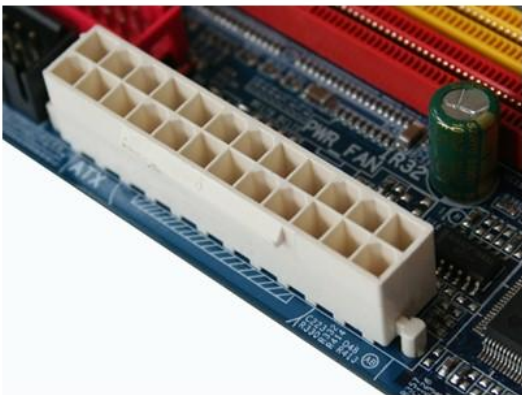
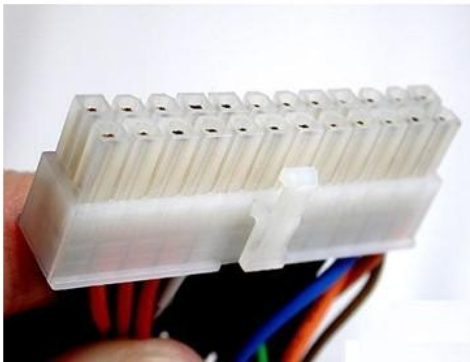
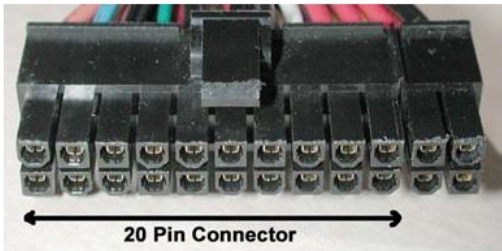
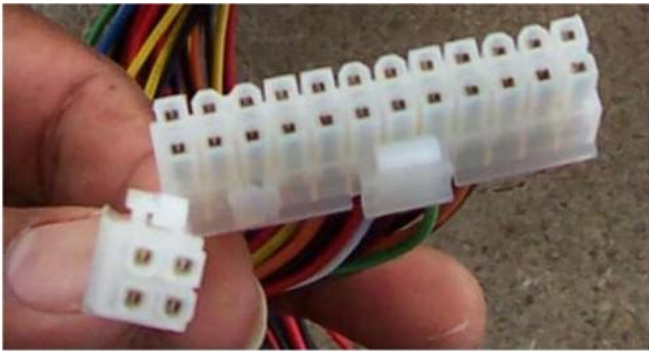
Large Molex Connectors



Berg Connector



ATX 24-pin Power Cable Connector



ATX 24-pin Power Cable Pinout

24 Pin ATX Pinout					
Pins 1 through 12			Pins 13 through 24		
Description	Wire color	Pin number	Pin number	Wire color	Description
+3.3 volts	orange	1	13	orange	+3.3 volts
+3.3 volts	orange	2	14	blue	-12 volts
ground	black	3	15	black	ground
+5 volts	red	4	16	green	PS_ON#
ground	black	5	17	black	ground
+5 volts	red	6	18	black	ground
ground	black	7	19	black	ground
PWR_OK	gray	8	20	white	-5 volts (optional)
VS5 +5 volts	purple	9	21	red	+5 volts
+12 volts	yellow	10	22	red	+5 volts
+12 volts	yellow	11	23	red	+5 volts
+3.3 volts	orange	12	24	black	ground

Serial (Serial Advanced Technology Attachment)ATA Power Cable



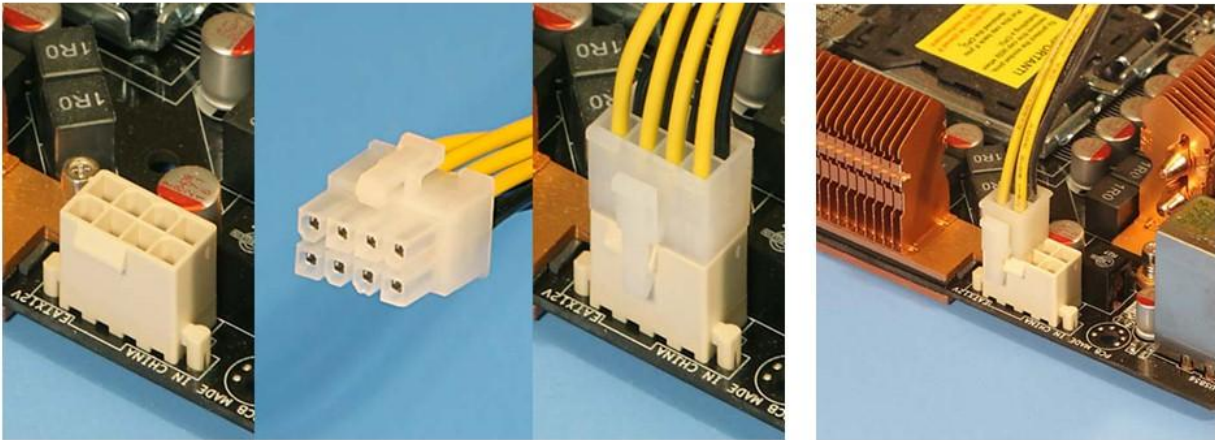
- 15-pin SATA Power Cable
- The power cable replaces the old 4 pin peripheral cable and adds support for 3.3 volts



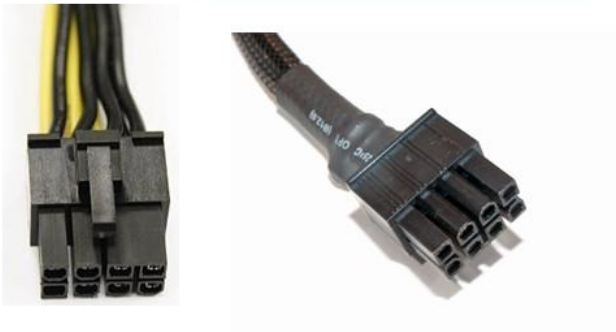
SATA Power Cable

Pinout			
Pin number	Wire number	Wire color	Description
1	5	orange	+3.3 volts
2	5	orange	+3.3 volts
3	5	orange	+3.3 volts
4	4	black	ground
5	4	black	ground
6	4	black	ground
7	3	red	+5 volts
8	3	red	+5 volts
9	3	red	+5 volts
10	2	black	ground
11	2	black	ground
12	2	black	ground
13	1	yellow	+12 volts
14	1	yellow	+12 volts
15	1	yellow	+12 volts

EPS 12V Connector



- **Entry-Level Power Supply Specification**
- is a Power Supply Unit meant for computers and entry-level servers.
- Developed by the Server System Infrastructure



EPS 12V Connector PINOUT

Pinout					
Pins 1 through 4			Pins 5 through 8		
Description	Wire color	Pin number	Pin number	Wire color	Description
ground	black	1	5	yellow	+12 volts (12V1)
ground	black	2	6	yellow	+12 volts (12V1)
ground	black	3	7	yellow	+12 volts (12V1 or 12V2)
ground	black	4	8	yellow	+12 volts (12V1 or 12V2)

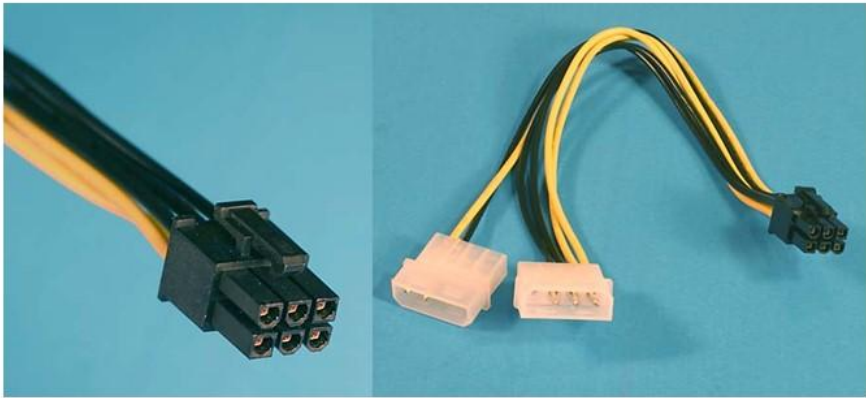


6 pin Power Connector

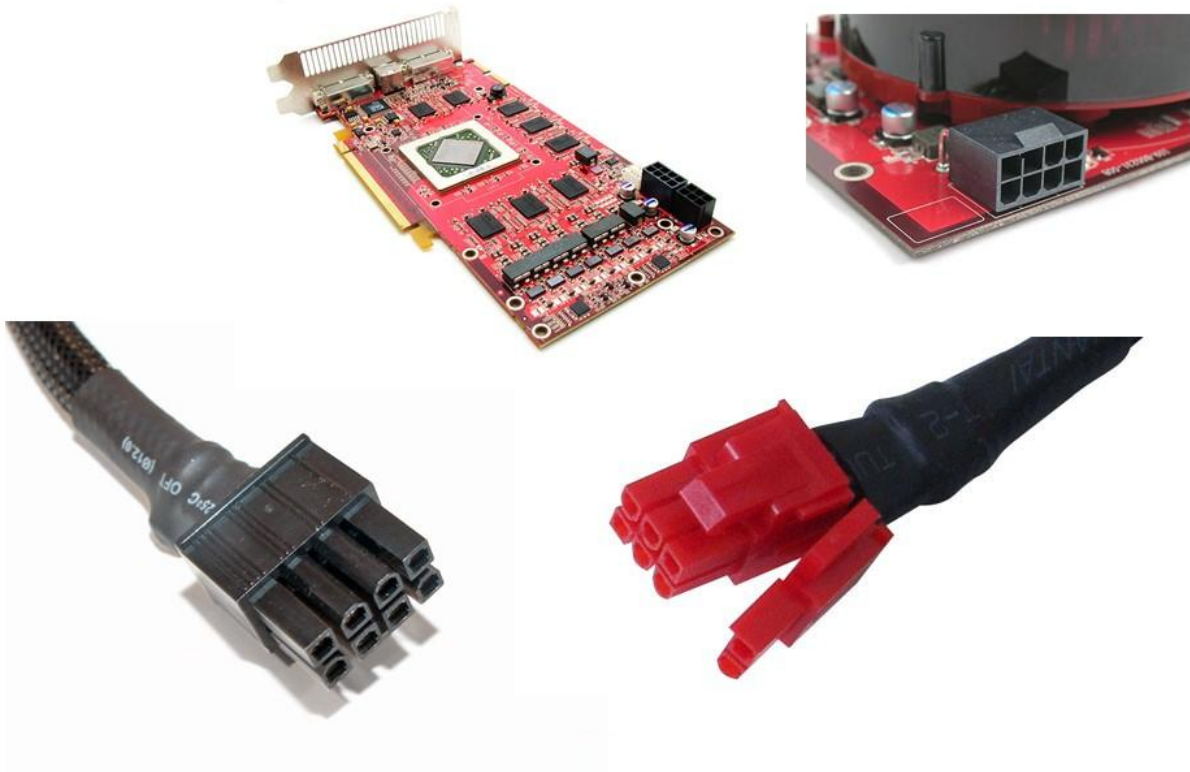


6 pin Power Connector Pinout

Pinout					
Pins 1 through 3			Pins 4 through 6		
Description	Wire color	Pin number	Pin number	Wire color	Description
+12 volts	yellow	1	4	black	ground
+12 volts or not connected	yellow or not connected	2	5	black	ground
+12 volts	yellow	3	6	black	ground



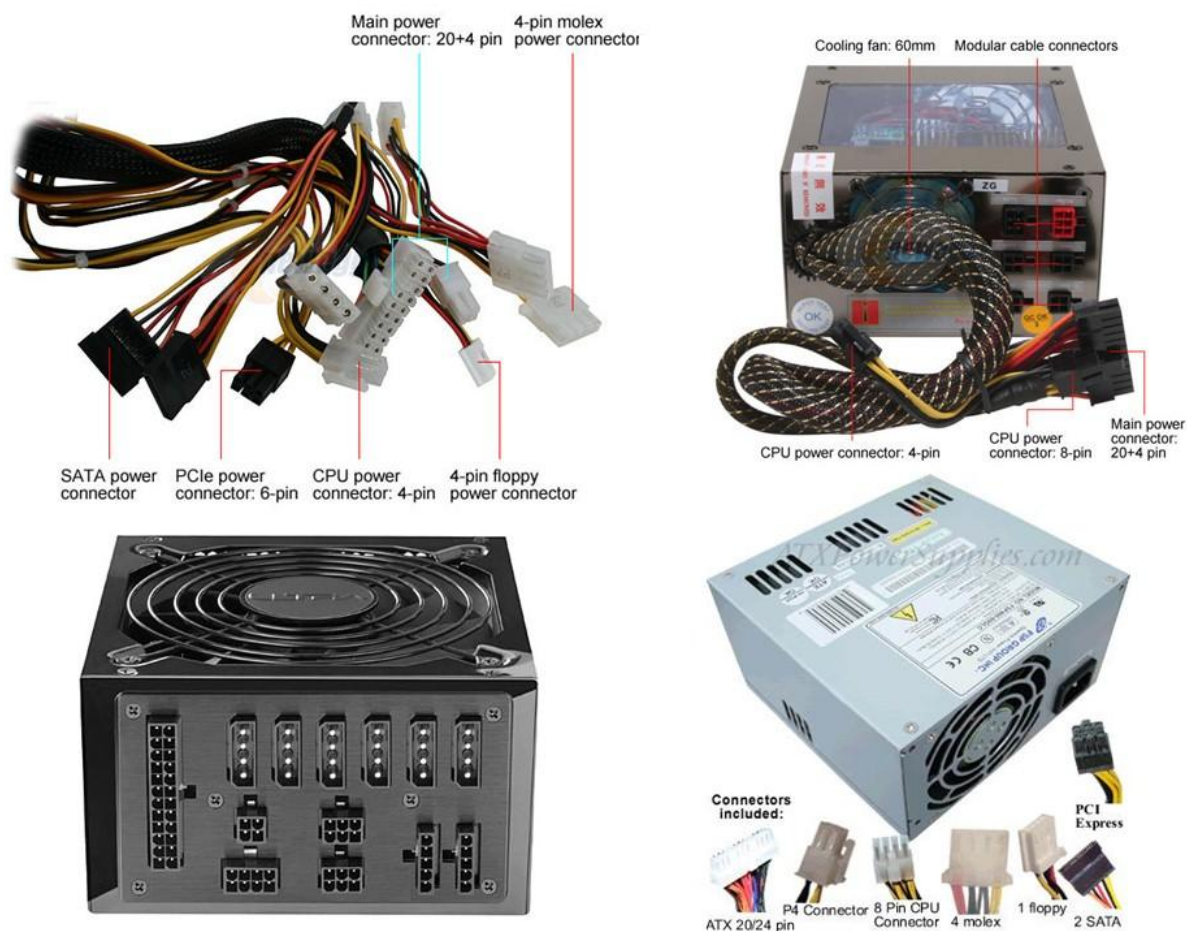
8 pin Power Connector



8 pin Power Connector Pinout

Pinout					
Pins 1 through 3			Pins 4 through 6		
Description	Wire color	Pin number	Pin number	Wire color	Description
+12 volts	yellow	1	5	black	ground
+12 volts	yellow	2	6	black	ground
+12 volts	yellow	3	7	black	ground
ground	black	4	8	black	ground



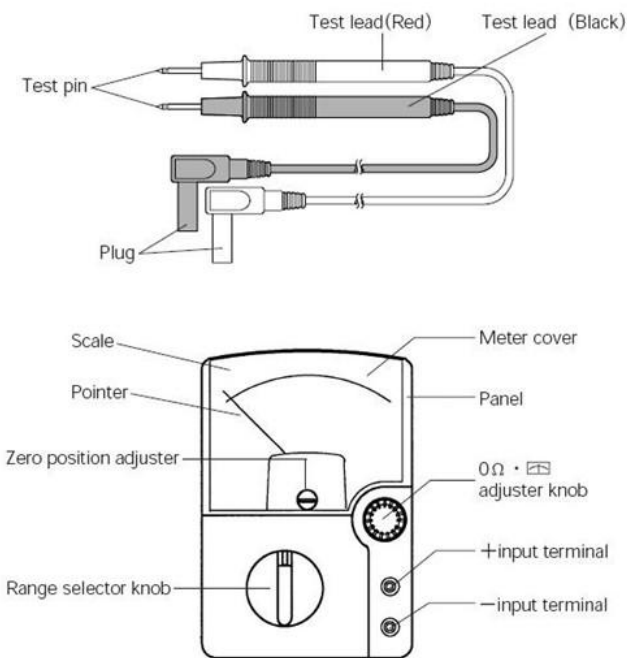


What is a Multitester:

- also known as a **volt/ohm meter** or **VOM**
- is a tool that can be use to check resistance, continuity, and voltage
- used to troubleshoot electrical problems in a wide array of industrial and household devices such as batteries, motor controls, appliances, power supplies, and wiring systems
- Digital / Analog



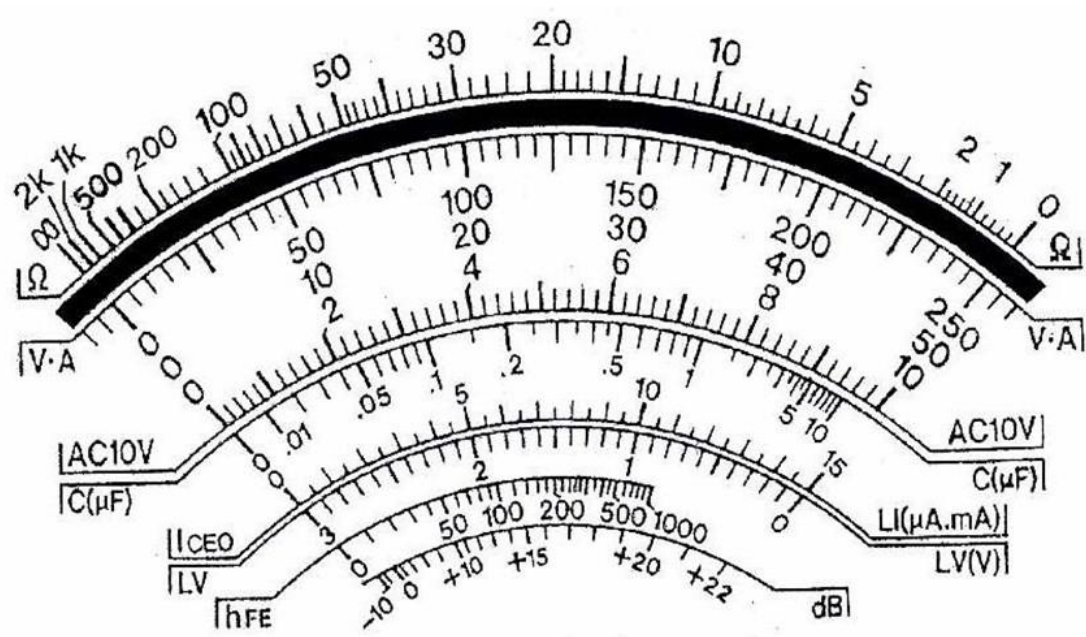
NAME OF COMPONENT UNITS



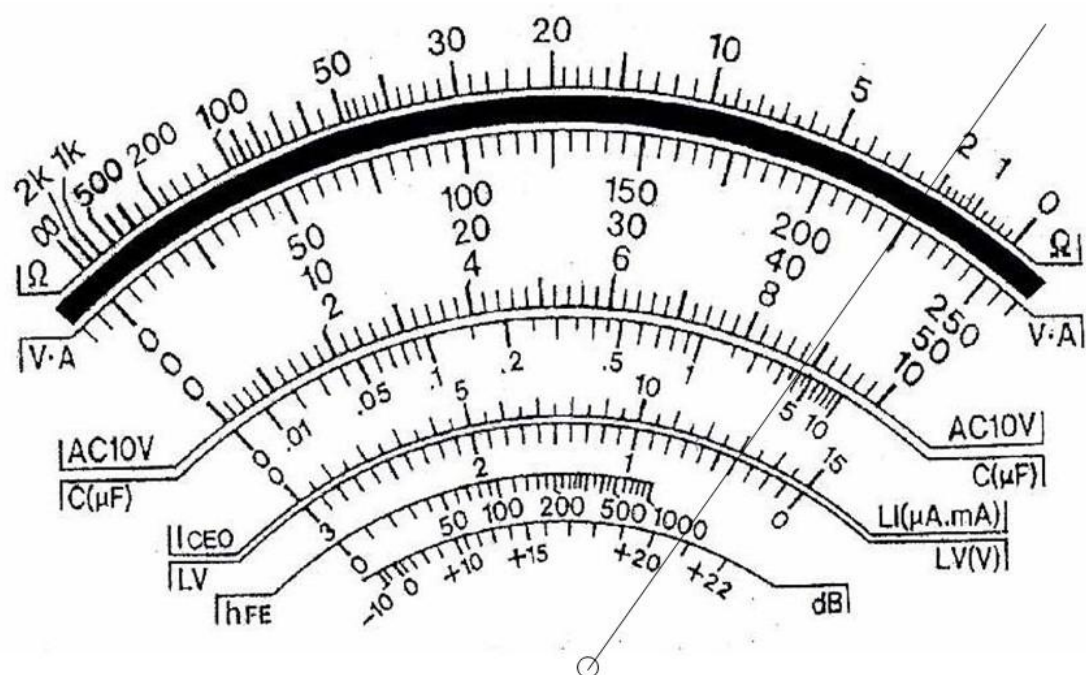
How to use a Multitester

- **How I check resistance or continuity:** To check for resistance or continuity you will need to set the multitester to red ohms.
- **How I check for voltage:** To check for voltage you need to set the multitester in to AC volts. My multitester has 3 settings 15 volts, 150 volts, and 1000 volts.
- **Why I do need different scales:** If you are checking an appliance that runs on 120 volts, then you can use the 150 scale. If you are checking an electric dryer that use 240 volts for the heating element, then you need to move the scale to higher that the voltage you are reading, in this case you need to set it to 1000 volts.

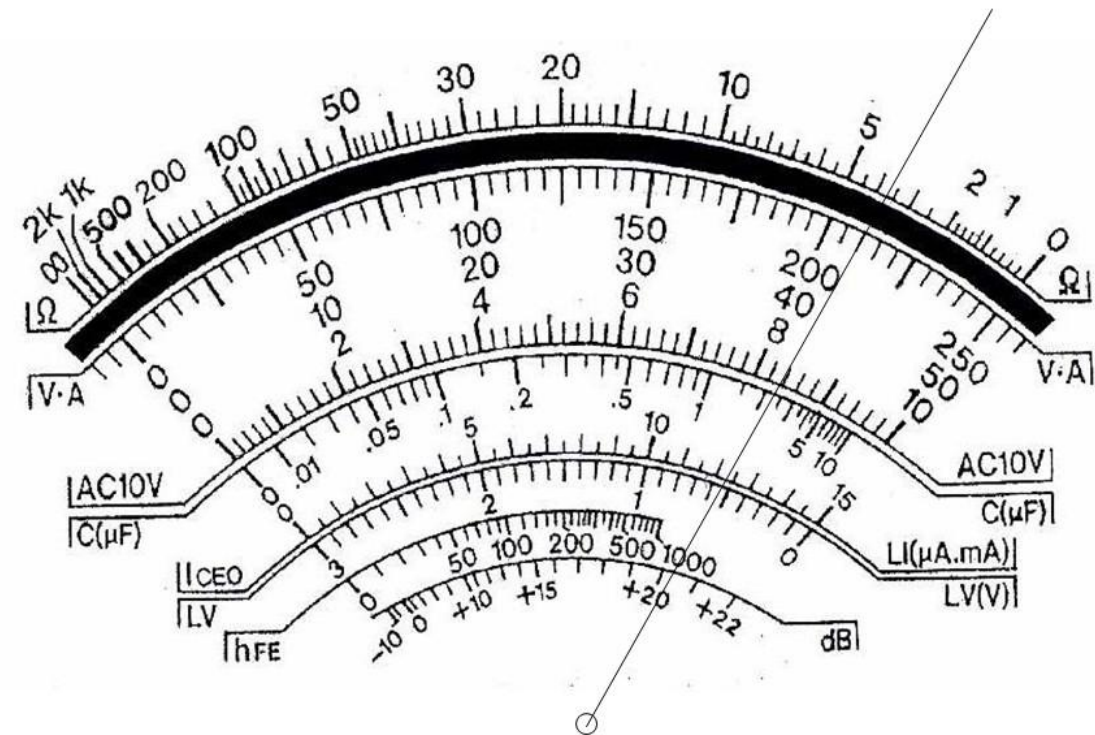




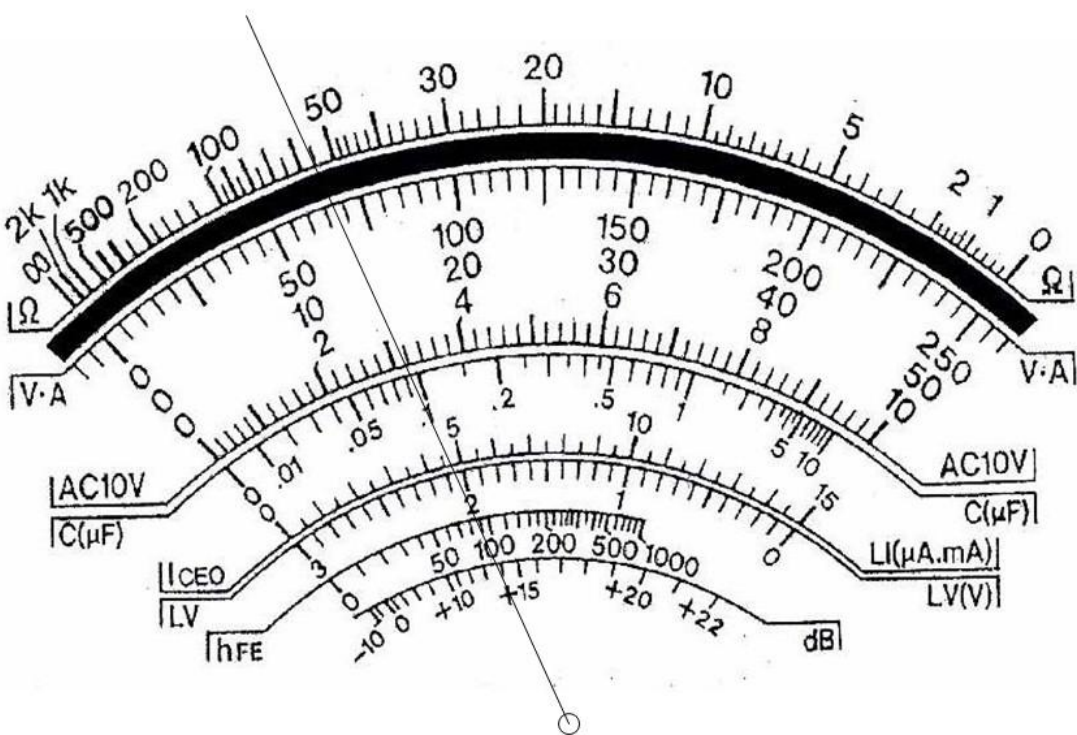
250 DC



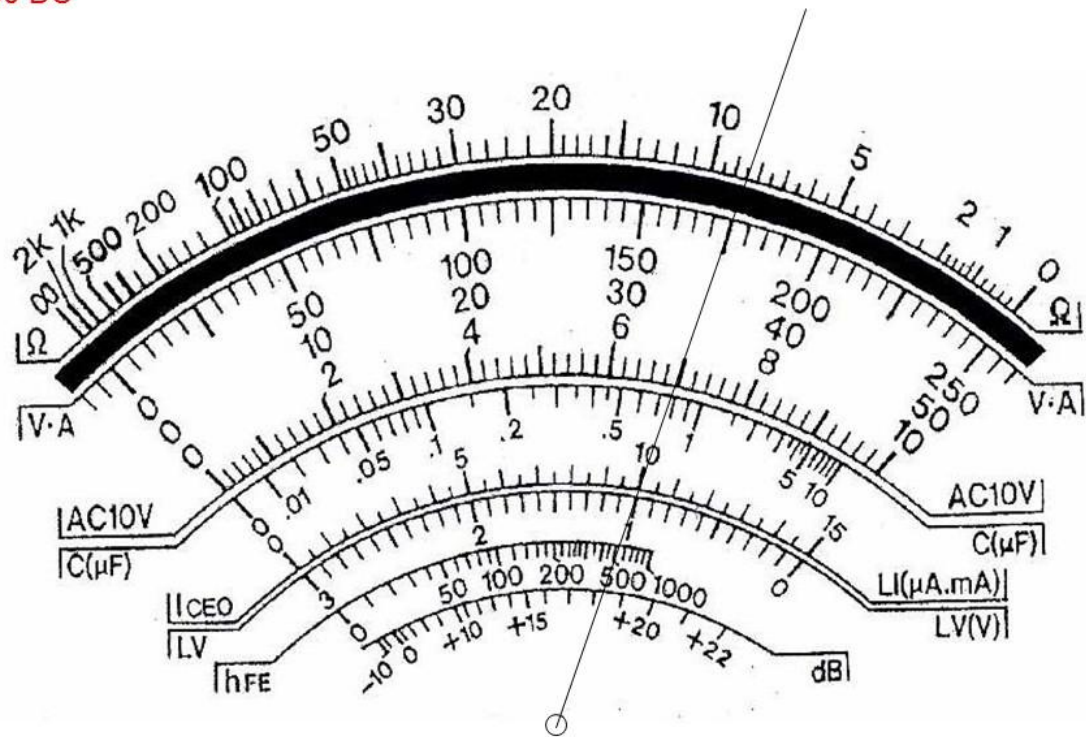
250 DC



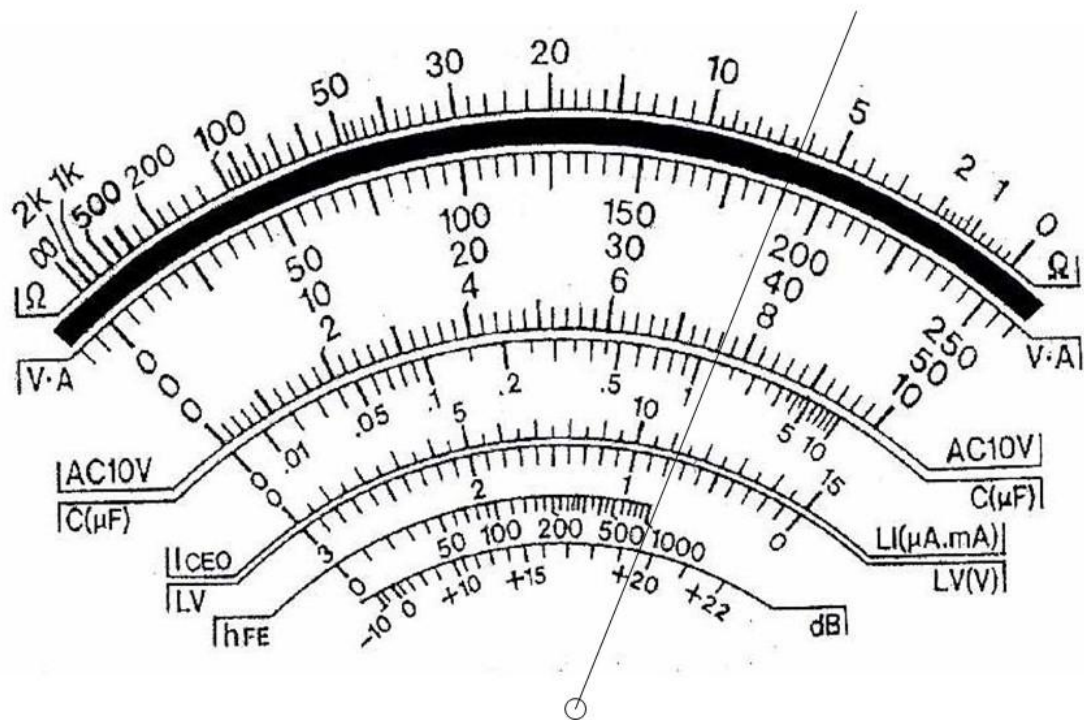
250 DC



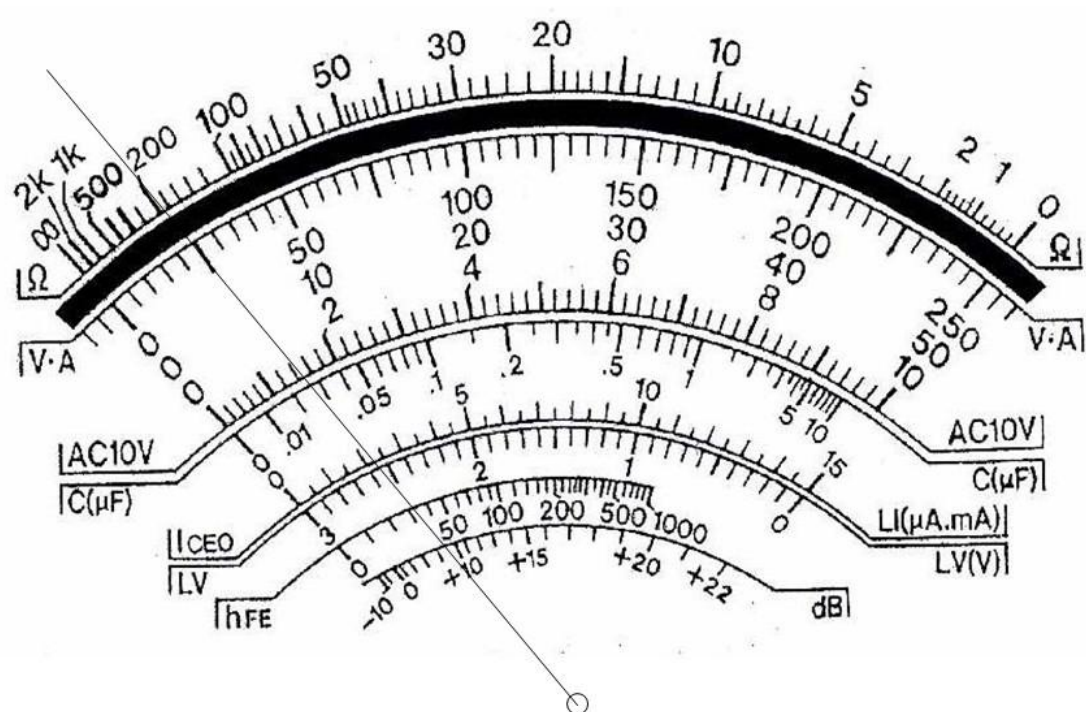
50 DC



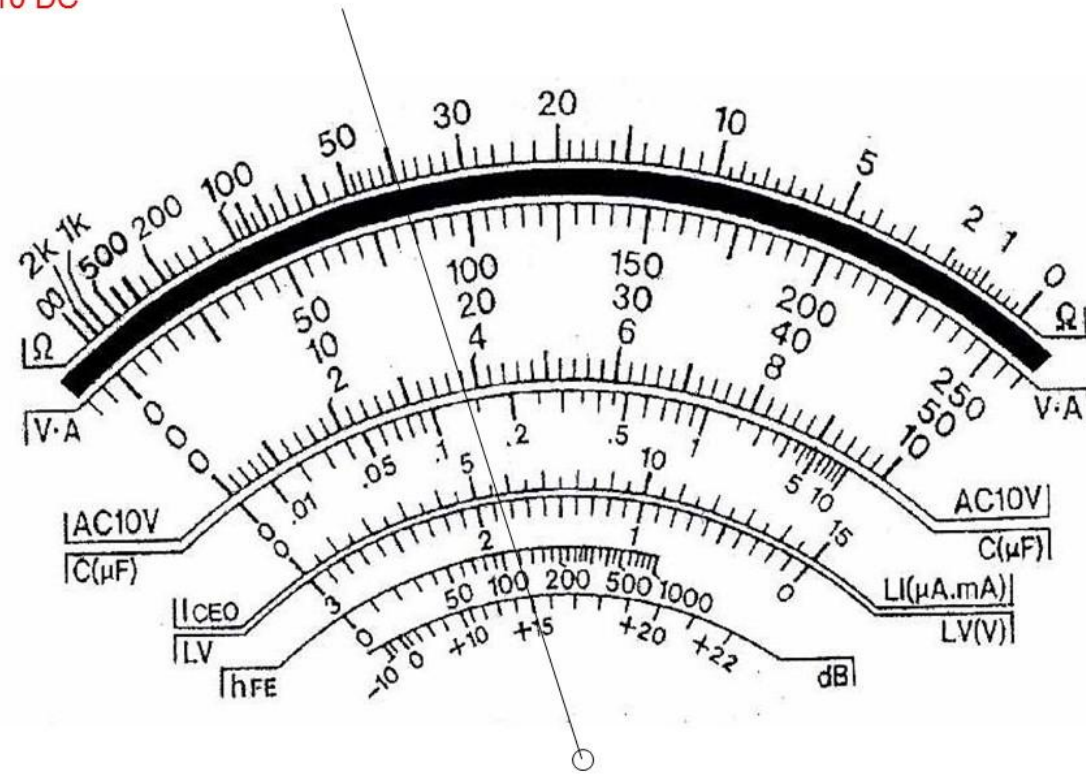
50 DC



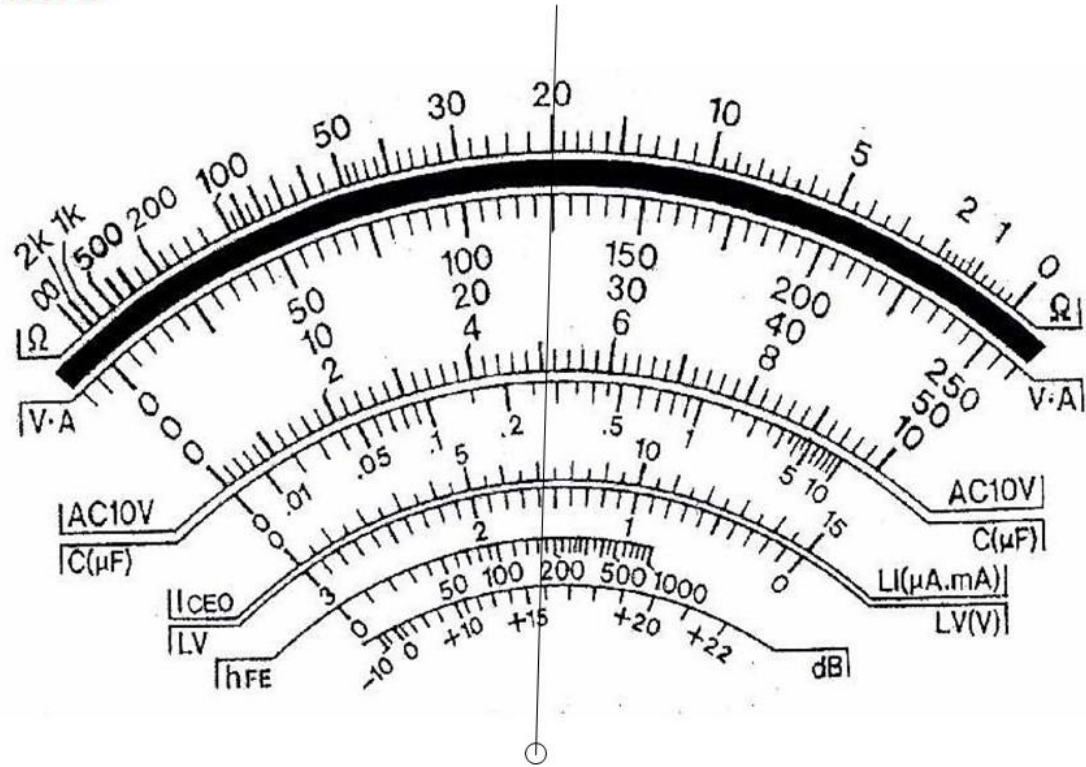
10 DC



10 DC



1000 AC



DISASSEMBLING THE PC

DISASSEMBLING THE PC

OH&S (Occupational Health & Safety)

- Observe ESD (Electrostatic Discharge)
- Always use an anti-static wrist strap
- Document everything
- Switch off the computer
- Remove the cables from the back of the case
- Be sure to have a container to keep the screws
- Place all components when remove in anti-static bags or anti-static mat
- Make sure you have all the tools,resource material, drivers or software that you need
- Do a full System Backup
- Use common sense

ESD

- *ElectroStatic Discharge* – is the transfer of an electrostatic charge between two objects. This is a very rapid event that happens when two objects of different potentials come into direct contact with each other.
- is one of the few things an individual can do to damage or destroy his or her computer or [hardware](#) components
- ESD can occur when working in your computer and will cause components you touch to no longer work properly.

Antistatic bag

- A bag that has been specially designed to help prevent static electricity inside the bag.
- used to store and transfer electrostatic sensitive hardware.
- It is important to remember that *only* the inside of an antistatic bag is protected, placing a hardware device on top of the bag and not inside of the bag will not protect it. In fact it is often more harmful to place a electrostatic sensitive device on top of the bag.



STATIC SHIELDING BAGS



Anti-static Bubble Bag

Antistatic Mat

- Also known as a **grounding mat**
- is a floor or table mat used to help reduce the risk of electrostatic discharge while working on a computer or other electrostatic sensitive object.



Wrist strap

- Also known as a **antistatic strap** or **antistatic wrist strap**
- The antistatic wrist strap prevents static electricity from building up. Static electricity can electronically destroy many of the delicate components inside the computer.
- An antistatic wrist strap should be worn any time work is being done inside the computer



How to help prevent ESD

- The best method of preventing ESD is to use an ESD wrist strap and/or use a grounding mat. However, because most users do not have access to such items, there are steps to help reduce the chances of ESD as much as possible.
1. **Zero Potential** - Most importantly, make sure you and the computer are at Zero Potential by continuously touching an un-painted metal surface of the chassis or the computer power supply case.
 2. **Standing** - It is also very important that you are standing at all times when working on the computer. Setting on a chair can generate more electrostatic.
 3. **Cords** - Make sure everything is removed from the back of the computer (power cord, mouse, keyboard, etc).
 4. **Weather** - Electrical storms can increase the ESD risk; unless absolutely necessary, try not to work on a computer during an electrical storm.
 5. **Accessories** - To help reduce ESD and help prevent other problem, it is also a good idea to remove all jewelry.

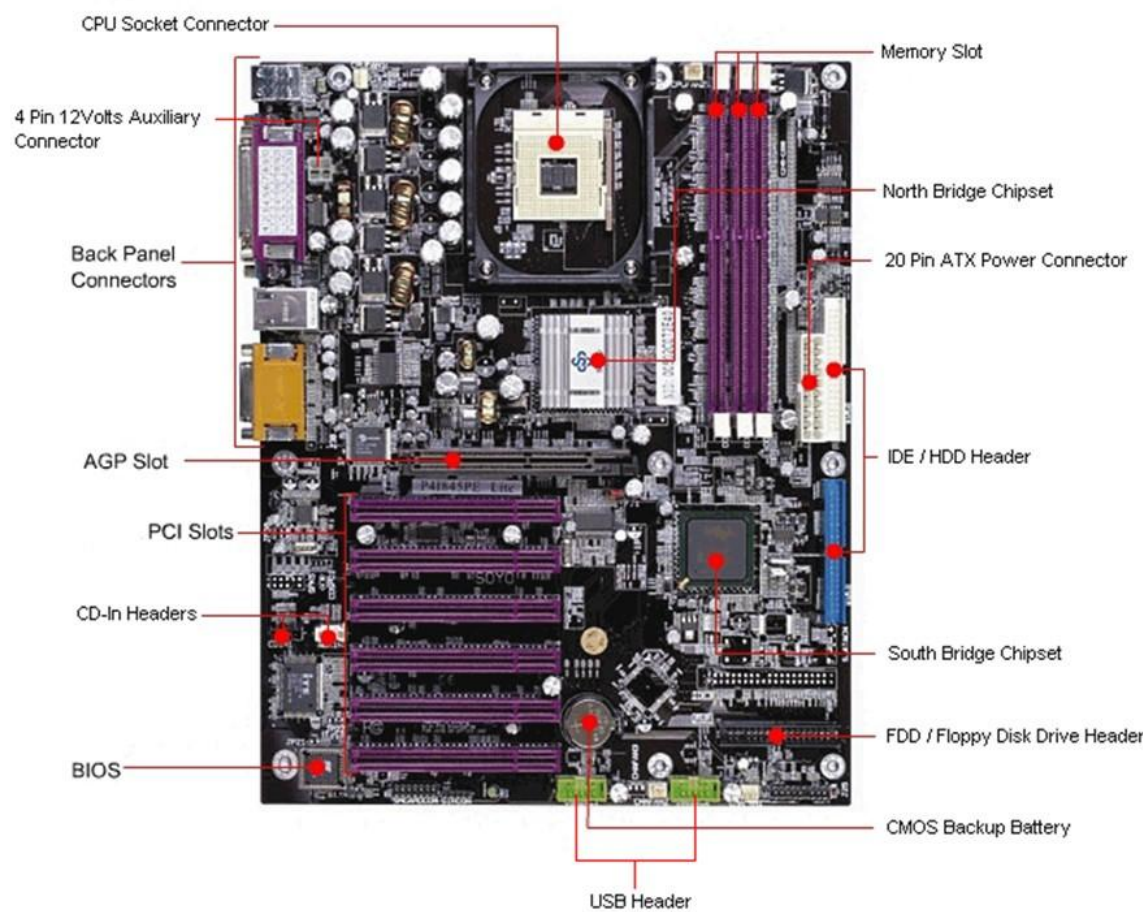
MOTHERBOARD

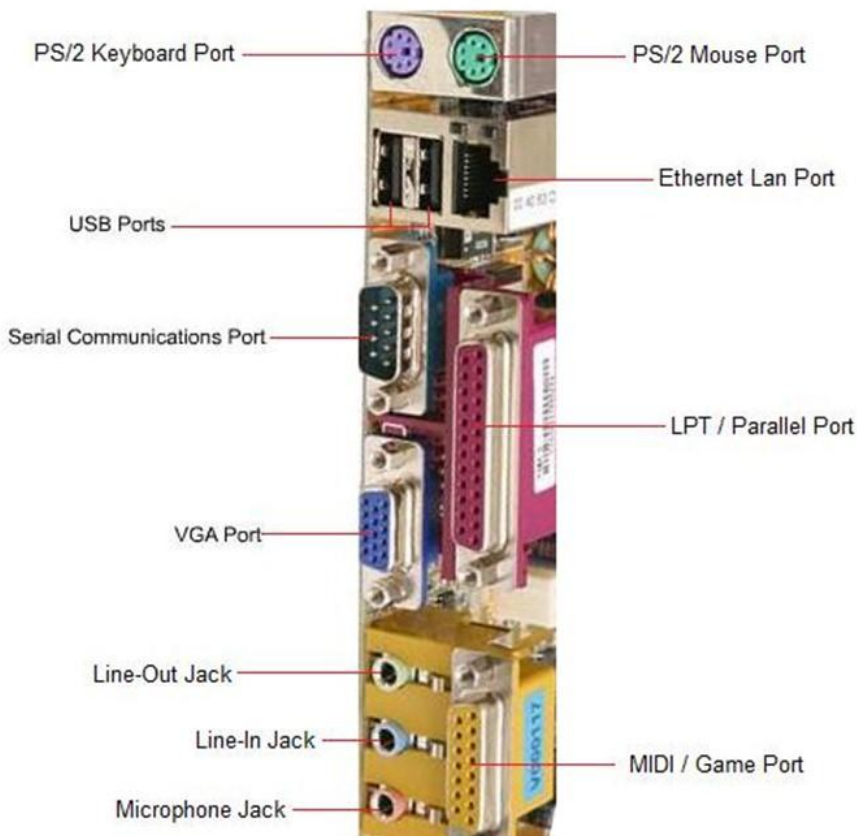
MOTHERBOARD

- It is also known as a mainboard, baseboard, system board, planar board or, on Apple computers, a logic board, and is sometimes abbreviated as mobo.
- it is the central or the main circuit board of your computer.
- All components and peripherals plug into it are place in there directly / indirectly
- and the job of the motherboard is to relay information between them all.



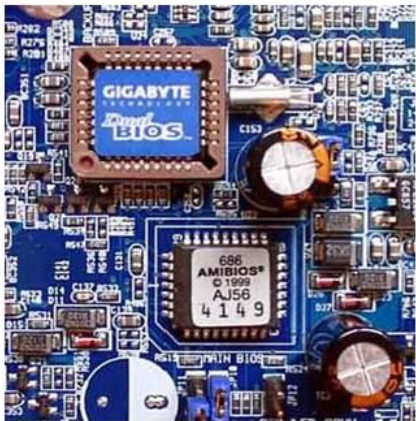
Parts of the Motherboard





BACK PANEL CONNECTORS

BIOS (Basic Input/Output System)



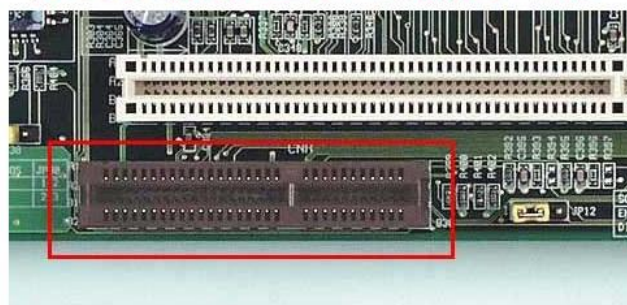
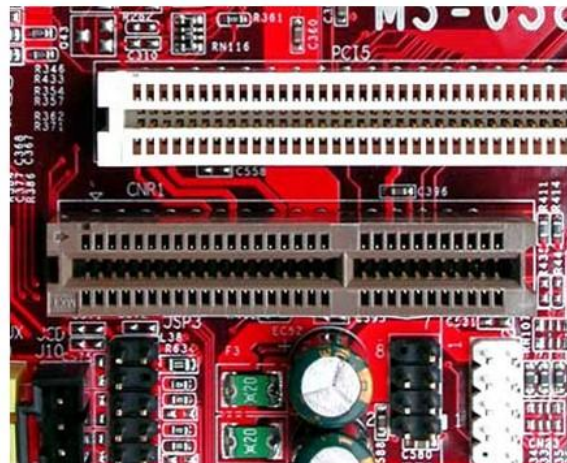
CMOS Battery

- Complementary metal–oxide–semiconductor
- refers to the memory on a personal computer motherboard containing BIOS settings and sometimes the code used to initialize the computer and load the operating system.
- The memory and real-time clock are generally powered by a CR2032 lithium coin cell.
- These cells last two to ten years, depending on the type of motherboard



CNR SLOT

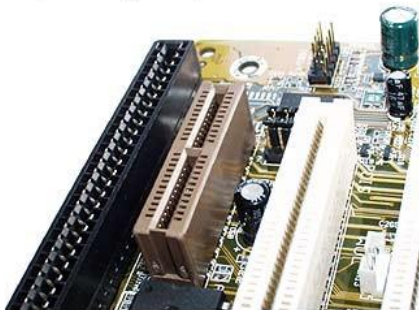
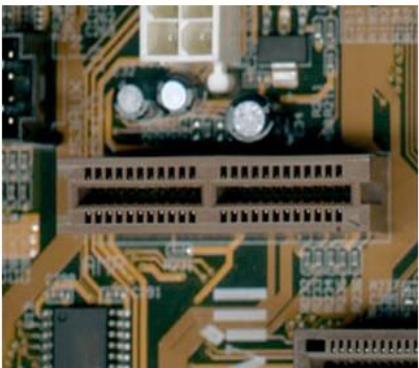
- *Communication and Networking Riser*
- used for specialized networking, audio, and telephony equipment.
- CNR slot has two rows of 30 pins



AMR Slot

Audio/modem riser

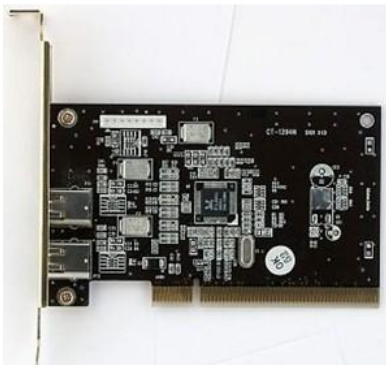
- also known as an **AMR slot**
- It was designed by [Intel](#) to interface with chipsets and provide analog functionality, such as [sound cards](#) and [modems](#), on an expansion card.
- Physically, it has two rows of 23 pins, making 46 pins total.



ACR

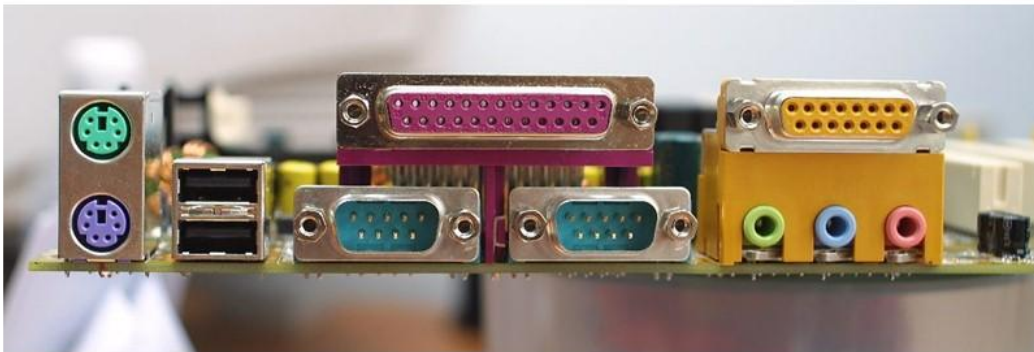
Advanced Communications Riser

- a replacement for Audio/modem_riser (**AMR**) slots, and a competitor and alternative to Communications and Networking Riser (**CNR**) slots.
- Used to connect certain expansion cards to a computer with an emphasis on audio and communications devices.
- uses a 120 pin PCI connector and has a backward compatibility with 46 pin AMR cards

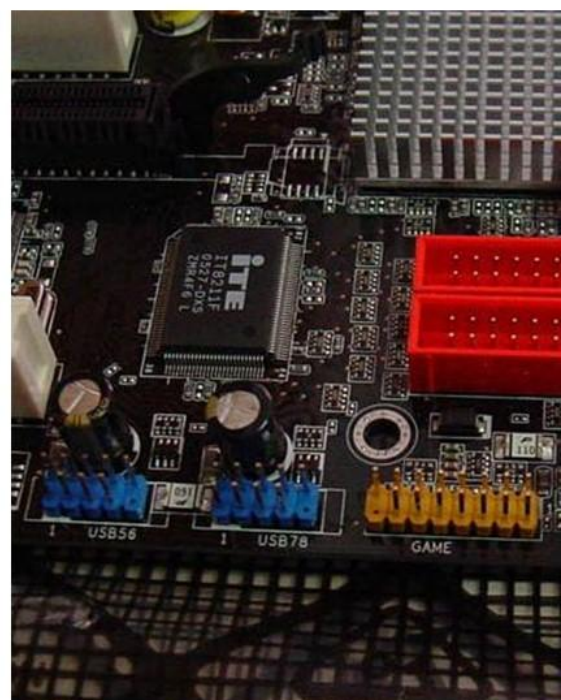
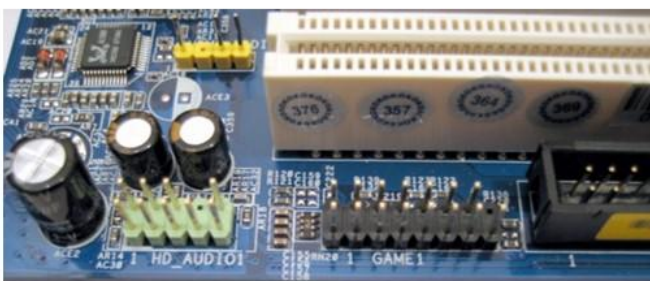


MIDI/GAME PORT

- **Musical Instrument Digital Interface**
- *is the traditional connector for [video game](#) input devices*
- either [ISA](#), Cable or [PCI](#)
- The 15-pin female gameport also acts as a MIDI port for [MIDI](#) instruments



GAME/MIDI PORT HEADER



24-Pin ATX Power Connector

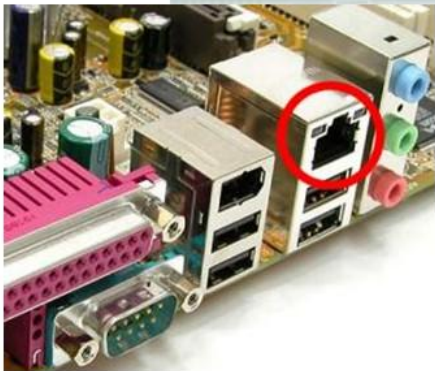


FIREWIRE PORT

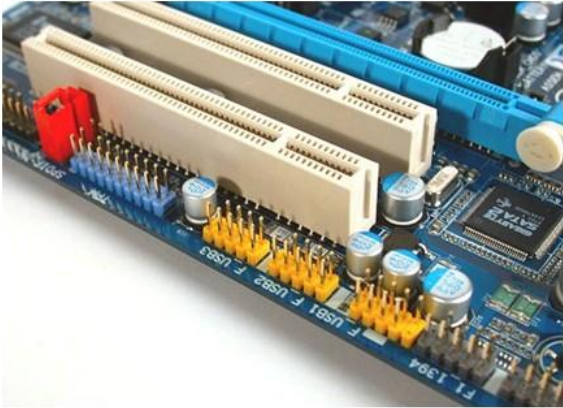
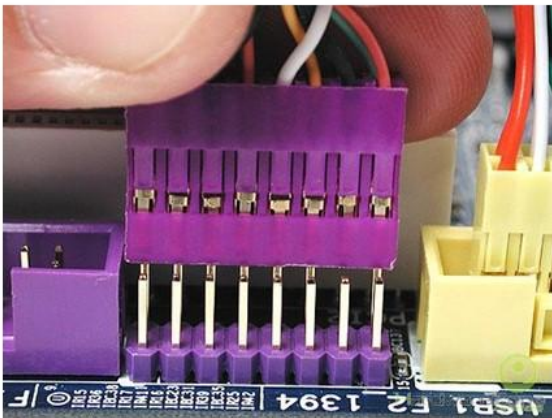


4-pin (left) and 6-pin (right)

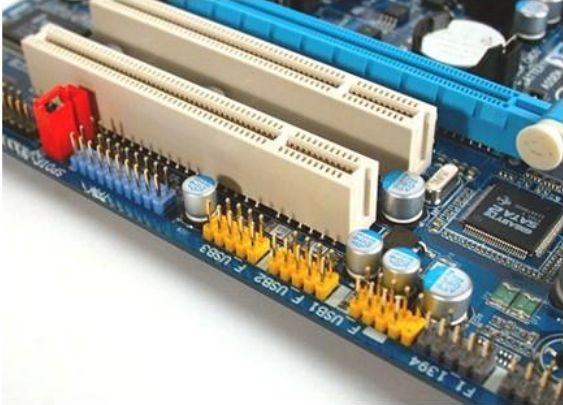
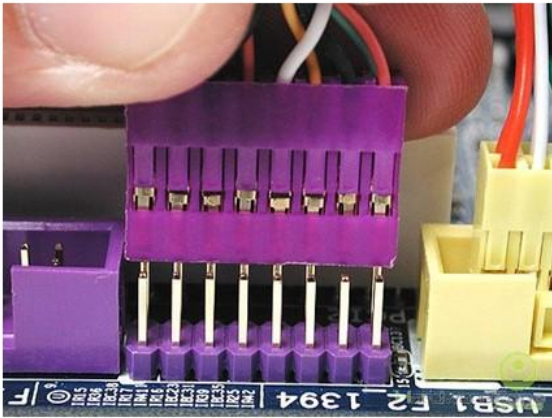
- FireWire is [Apple Inc.](#)'s name for the IEEE [1394](#) High Speed Serial Bus.
- is commonly used for connection of [data storage devices](#) and [DV](#) (digital video) cameras
- data transfer rates are higher for FireWire than for [USB 2.0](#)
- The 6-pin connector is commonly found on desktop computers



FIREWIRE HEADER



FIREWIRE HEADER



S/PDIF



- is an acronym for Sony /Philips Digital Interface (or Sony /Philips Digital Interconnect Format).
- is used for transporting stereo digital audio signals on PC audio cards, CD players, DVD players, car audio systems, and other systems which transmit or receive stereo digital audio.



RCA to mini-jack adapter



S-Video cable



RCA cable



Analog video capture device

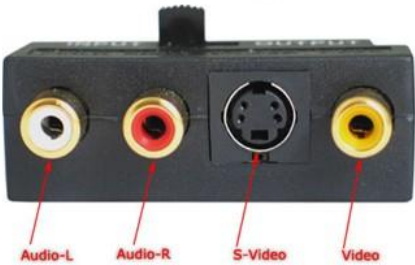
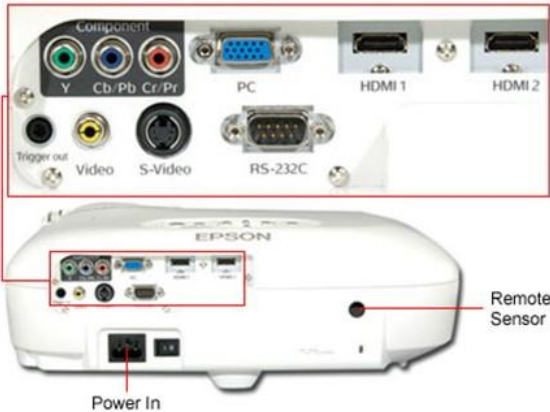


TV Tuner



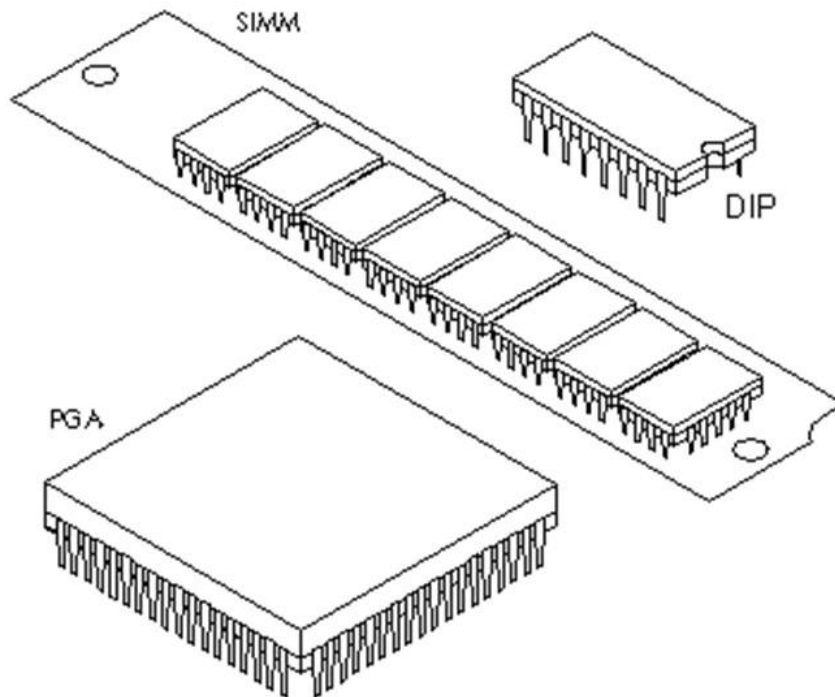
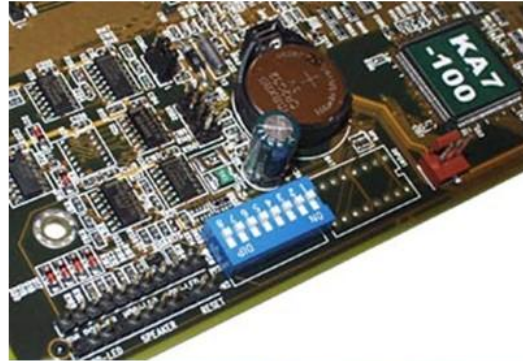
Composite video

- is the format of an analog television (picture only) signal
- It is usually in a standard format such as NTSC (National Television System Committee), PAL (Phase Alternating Line), or SECAM (Sequential Color with Memory).



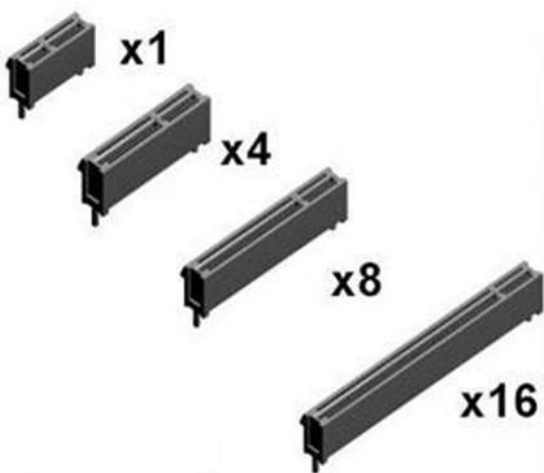
DIP Switch (dual in-line package)

- is designed to be used on a printed circuit board along with other electronic components and is commonly used to customize the behavior of an electronic device for specific situations.
- They were extensively used in older ISA PC cards to select IRQs (**interrupt request**) and memory addresses.
- DIP switches often come in packages of seven or eight.
- DIP switches are an alternative to jumper blocks. Their main advantages are that they are quicker to change and there are no parts to lose, although jumper blocks are more often used due to lower cost.



PCI EXPRESS SLOT

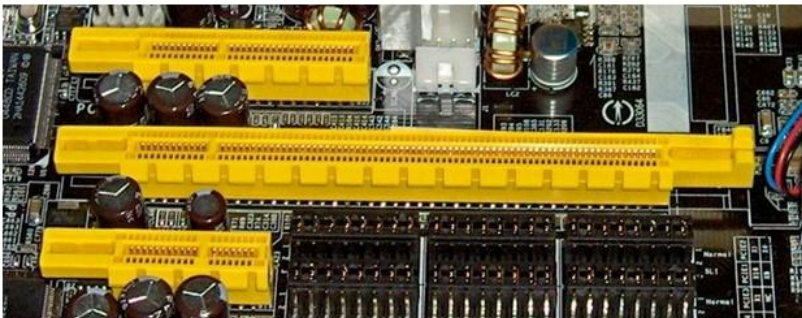
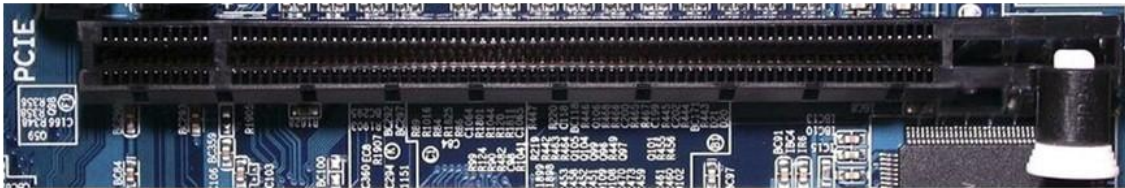
- officially abbreviated as **PCI-E** or **PCle**
- is a computer expansion card interface format introduced by Intel in 2004.



Lanes	Pins Total
x1	36
x4	64
x8	98
x16	164

- PCI Express was designed to replace the general-purpose [PCI](#) expansion bus

PCI EXPRESS SLOT



DVI Port

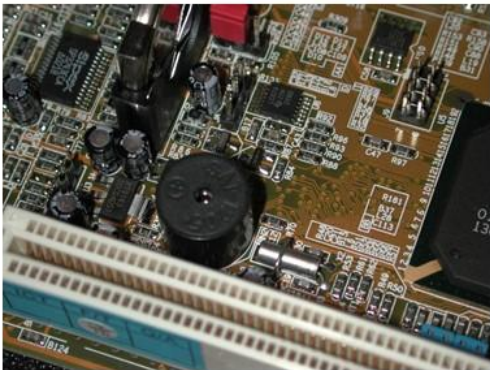
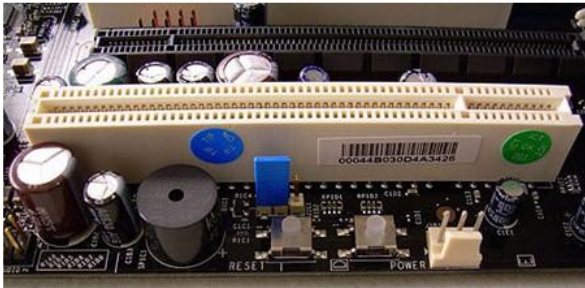
- Digital Visual Interface
- is a video interface standard designed to maximize the visual quality of digital display devices such as flat panel LCD computer displays and digital projectors.
- is a digital video standard introduced in 1999. It was developed by an industry consortium, the Digital Display Working Group (DDWG).
- Has 29 Pins



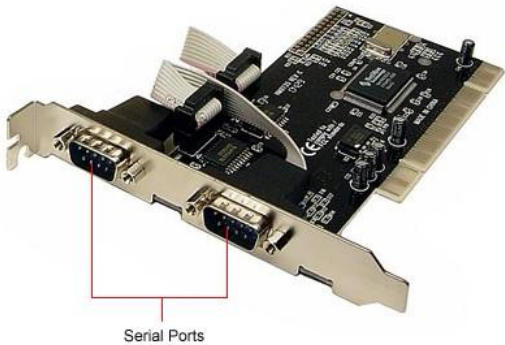
Converter



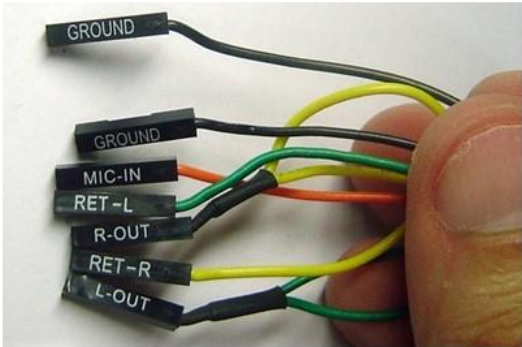
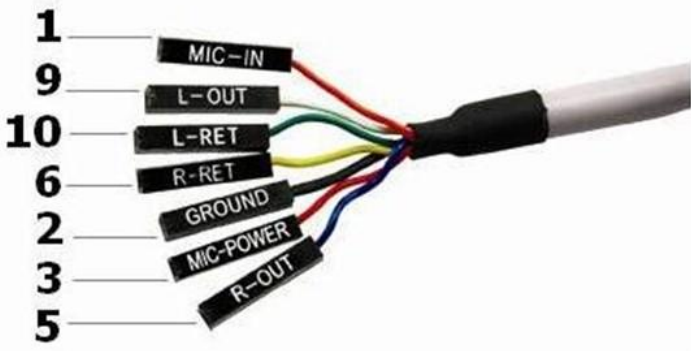
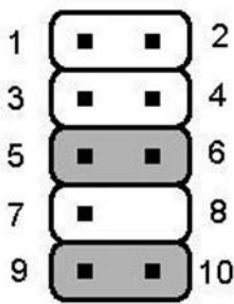
POST (power on self test) Speaker



Serial Communication Port

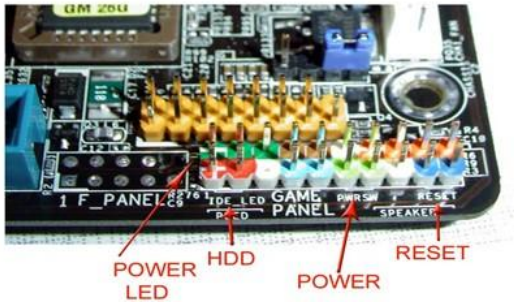
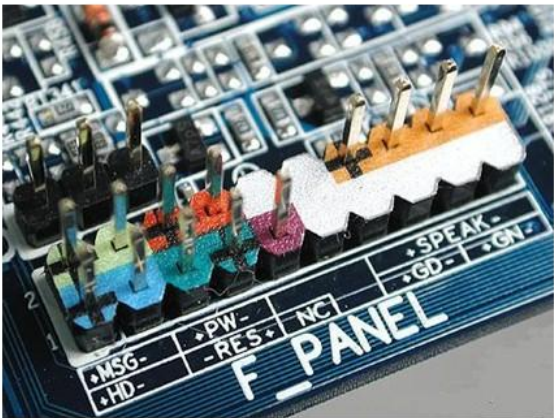
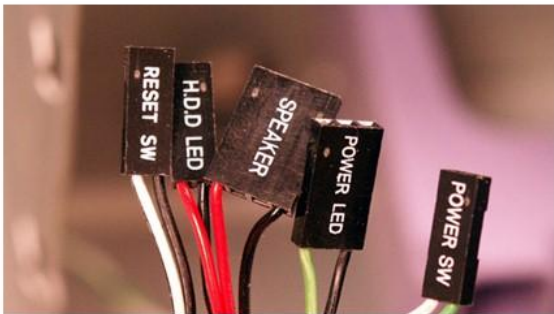


Analog Front Panel Audio Header

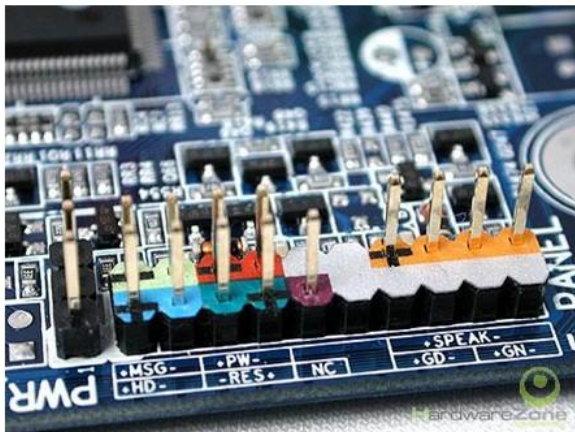


System / Front Panel Header

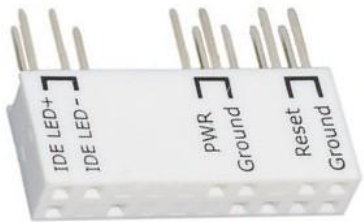
- consist of power switch, reset switch, power LED, hard drive activity LED and internal speaker connectors



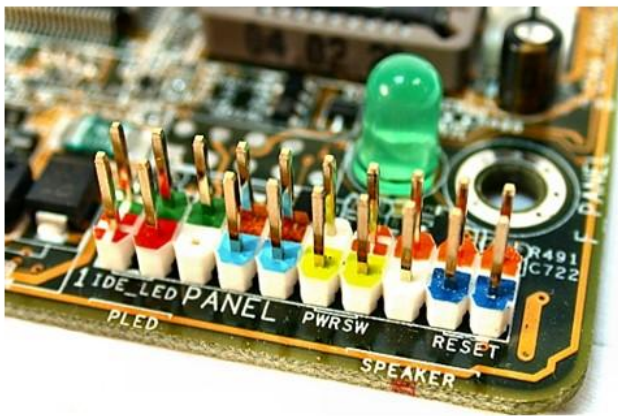
How to Install Front Panel Connectors?



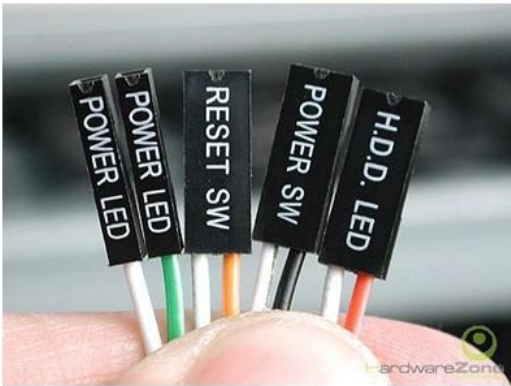
Front Panel Header

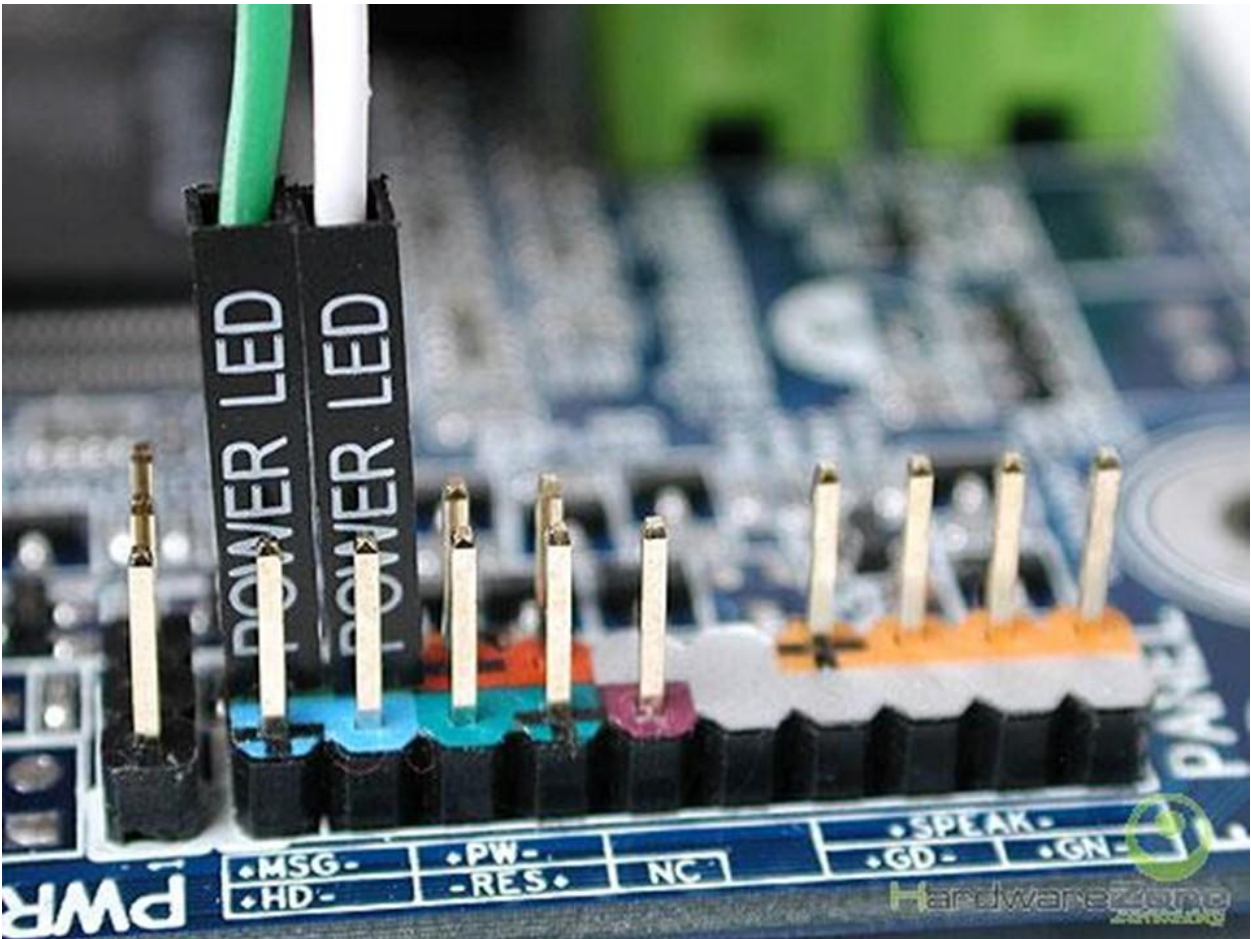
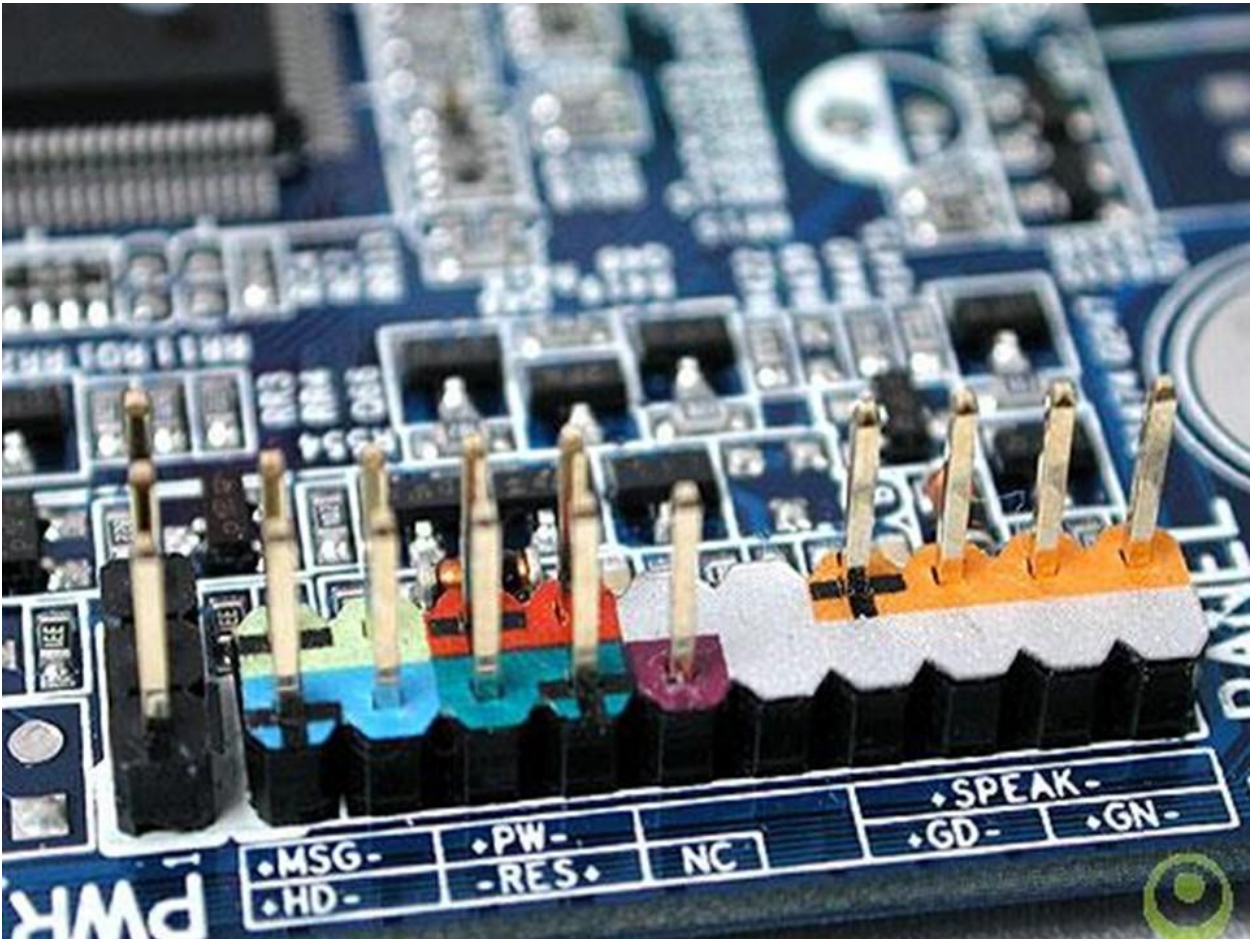


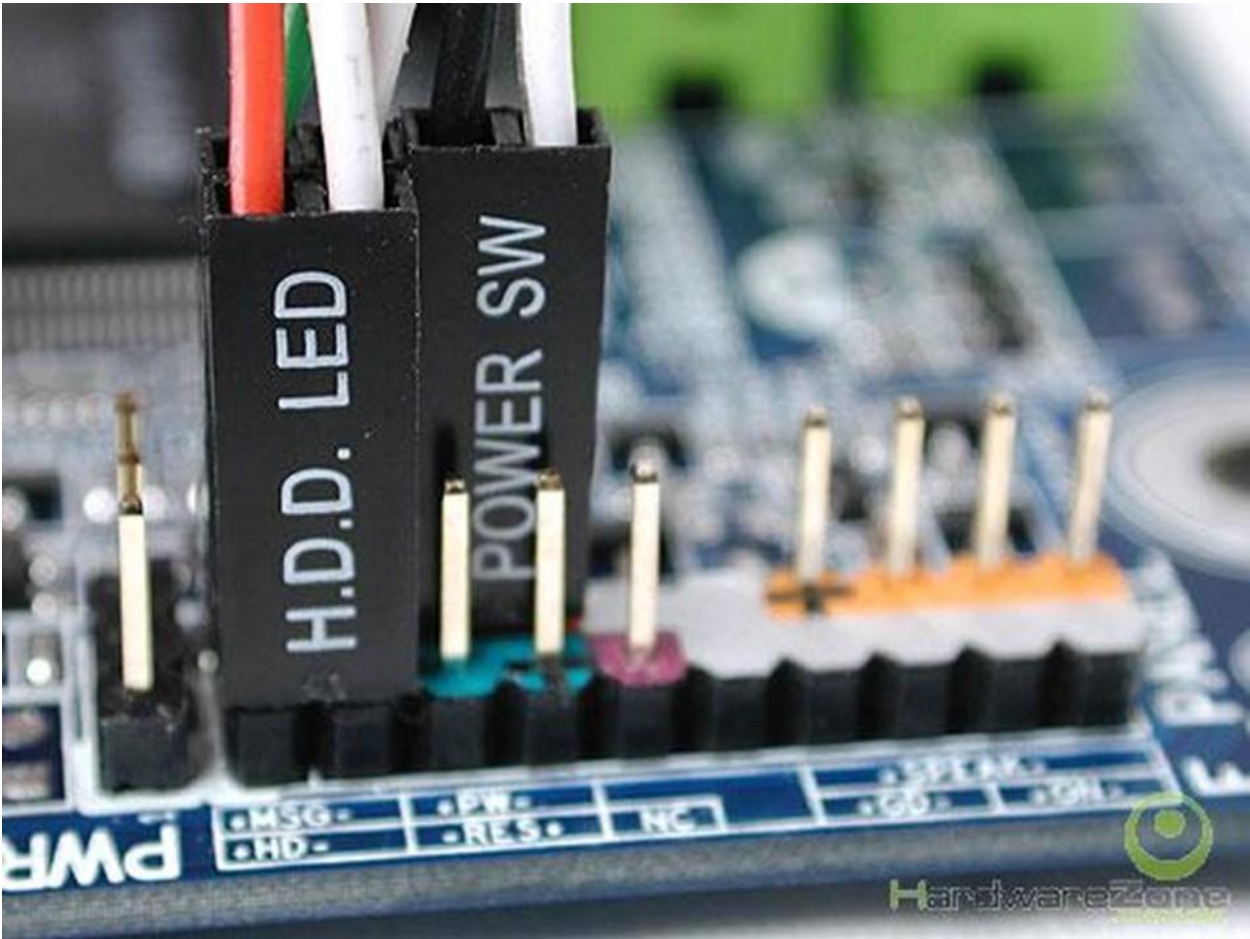
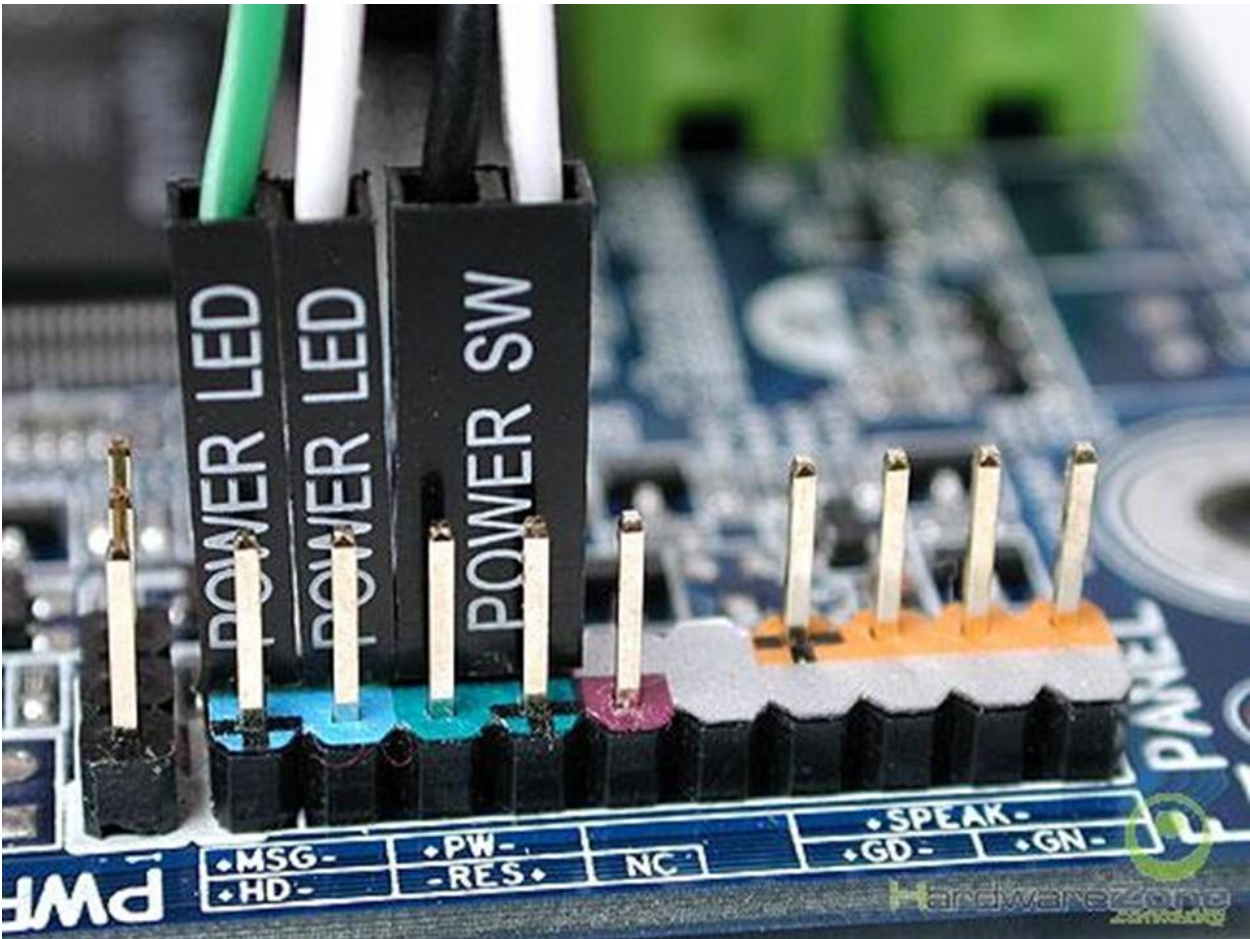
Q-Connector

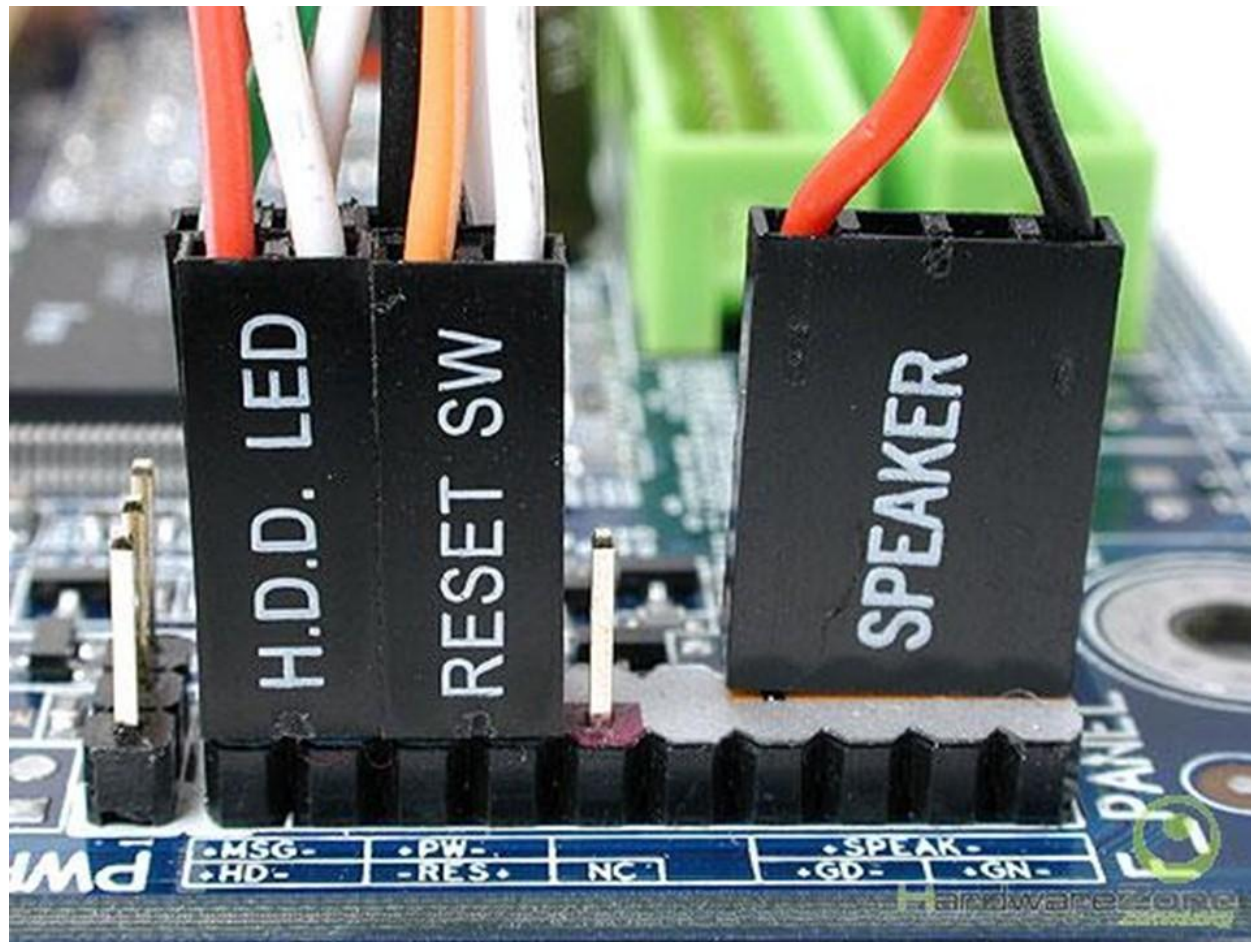
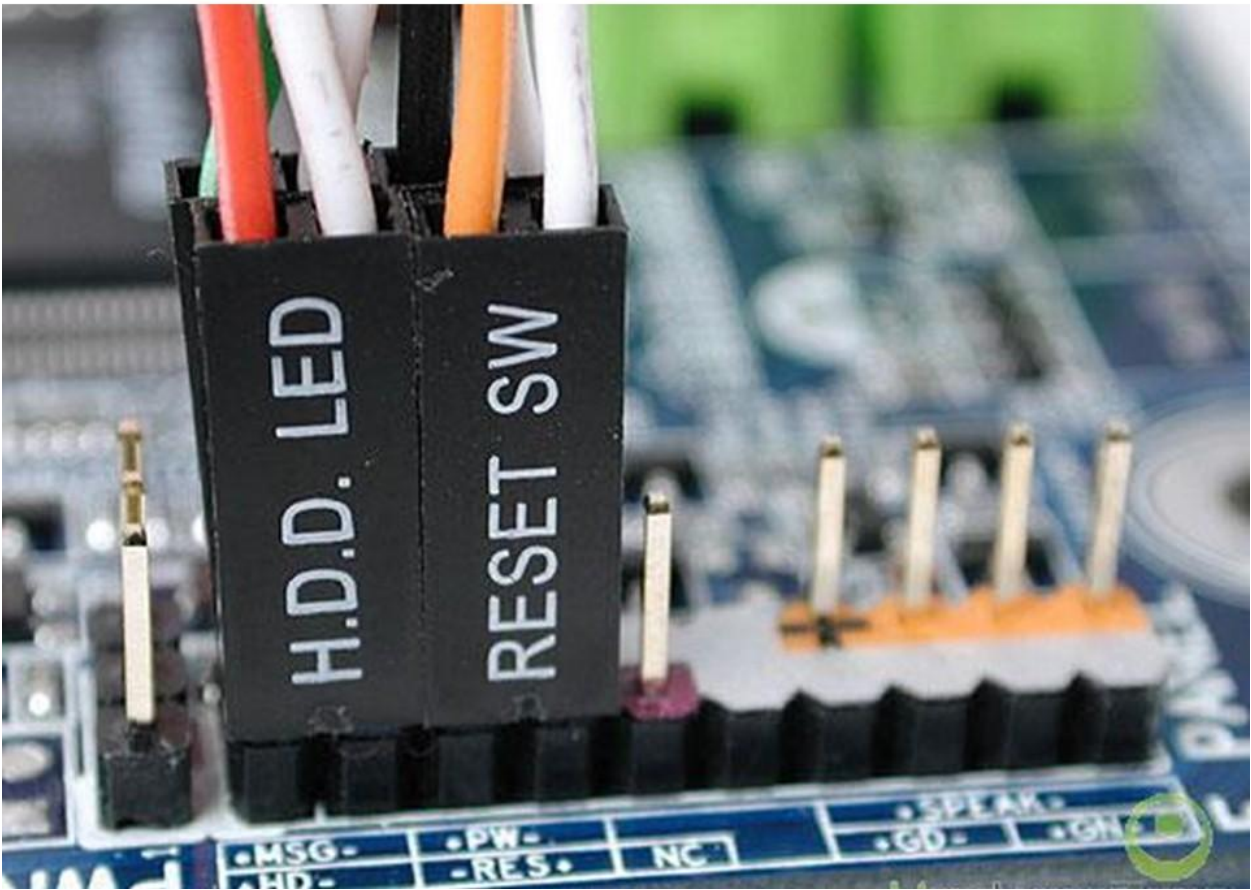


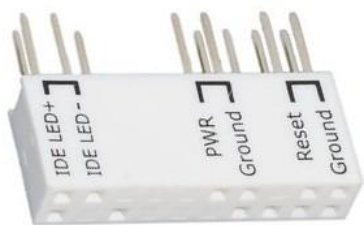
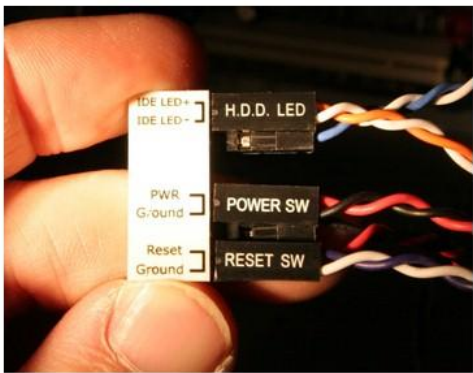
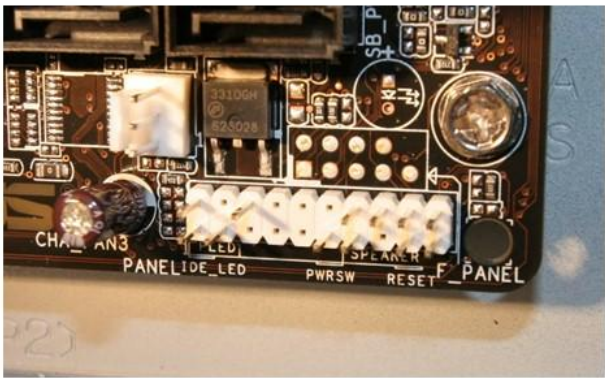
Front Panel Connectors



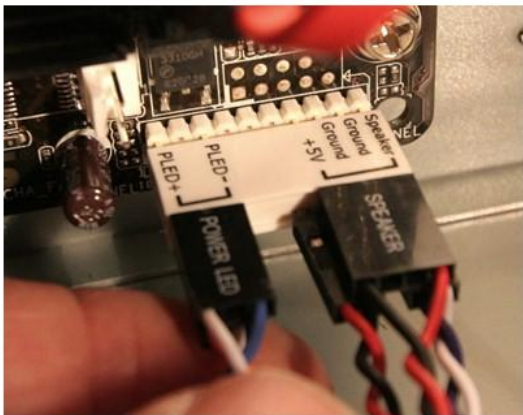




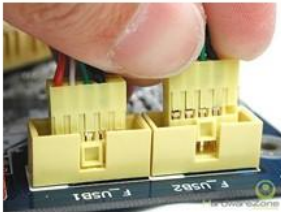
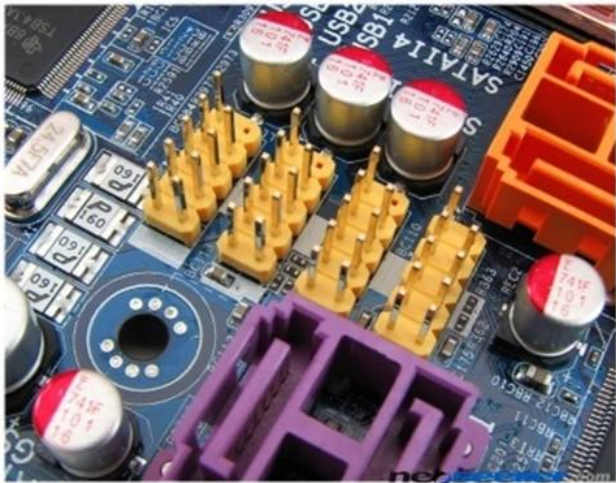




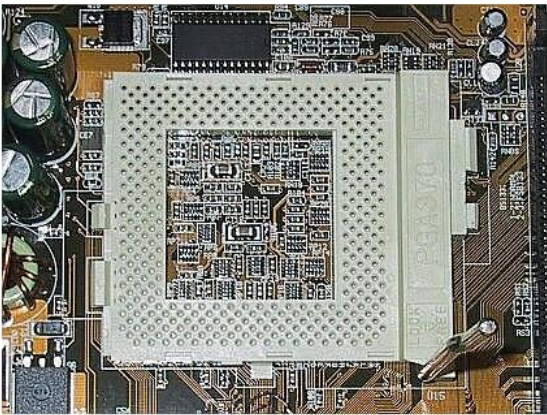
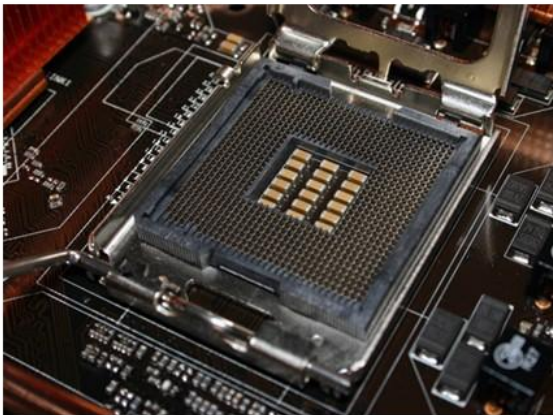
Q-Connector



USB Header



CPU Socket Connectors



PCI SLOTS

(Peripheral Component Interconnect)

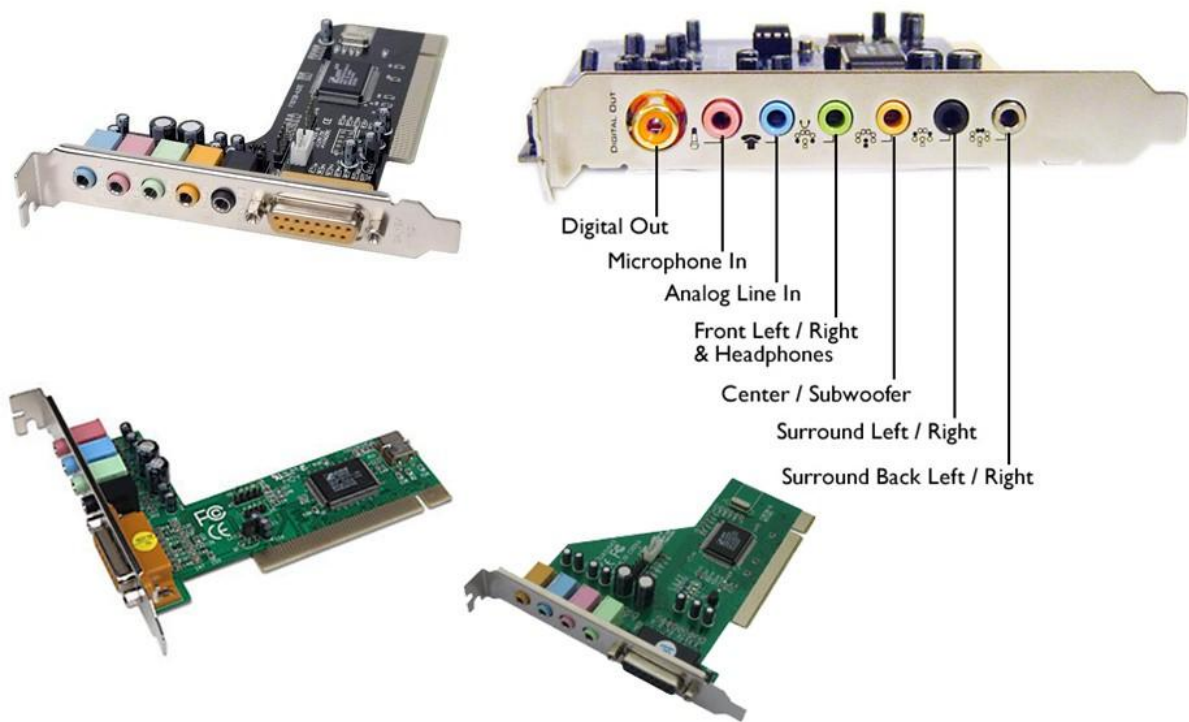


PCI DEVICES

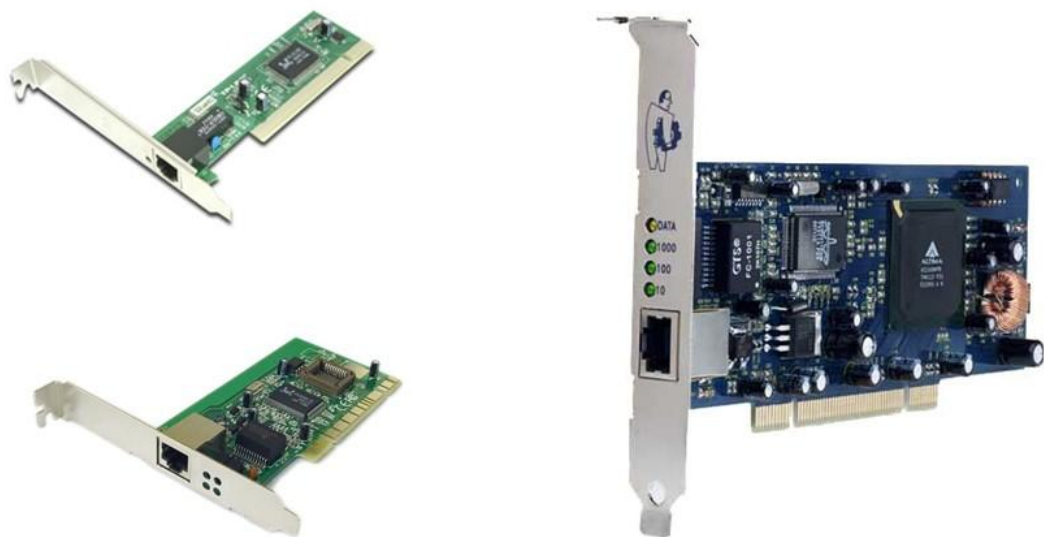
TV Tuner



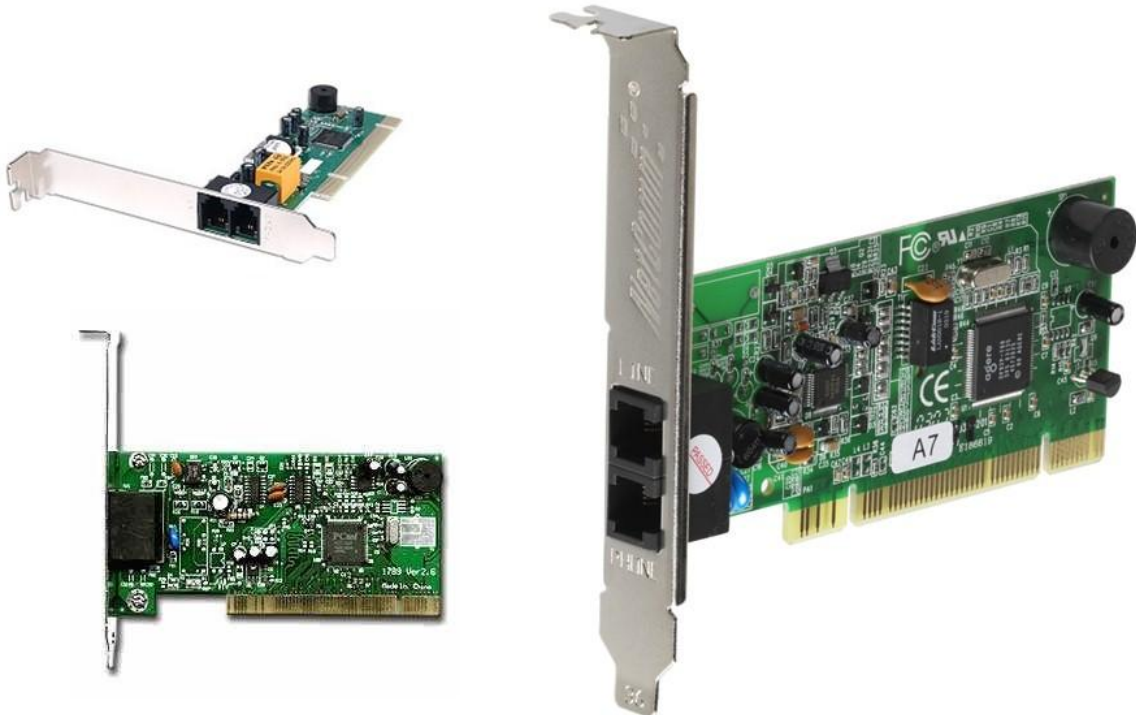
Sound Card



Network Cards



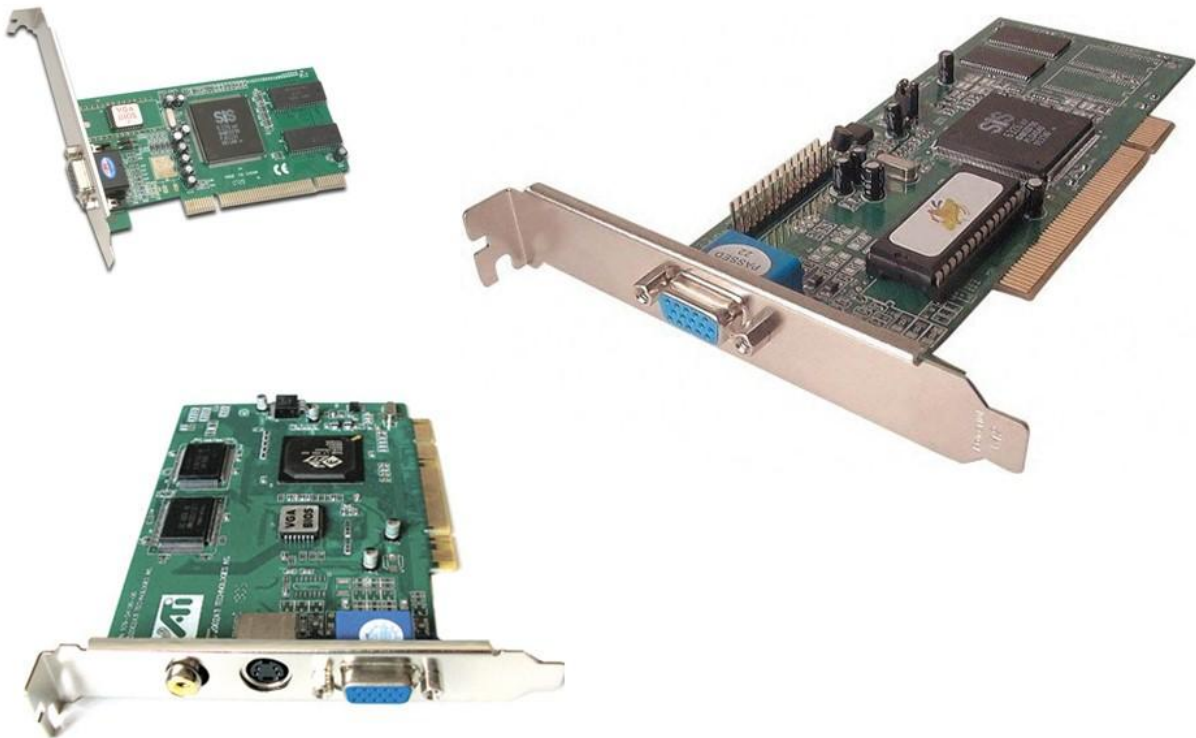
Modems



USB



Video Card



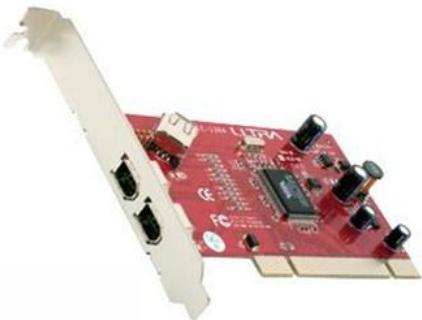
Wireless Network Adapters



SATA

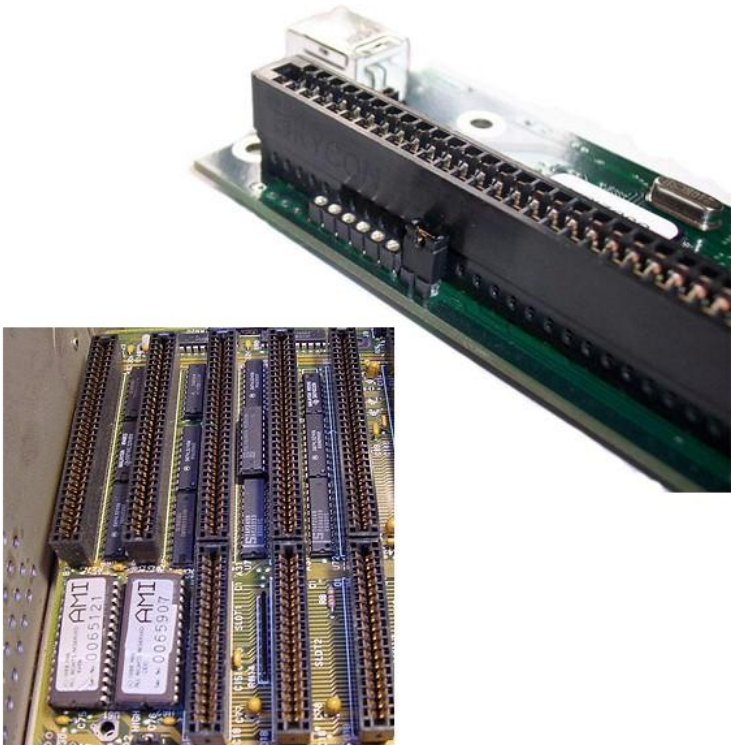


IEEE 1394 Firewire



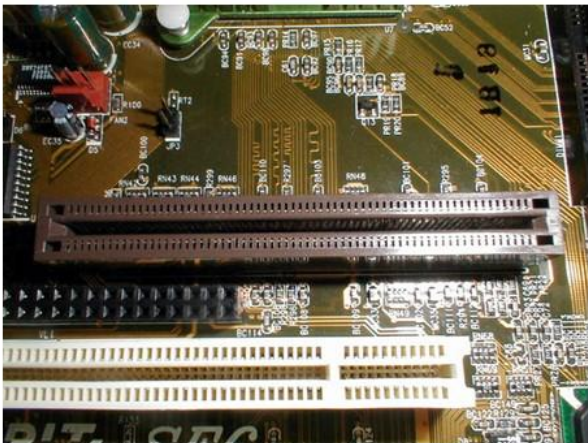
ISA Slot

- Industry Standard Architecture
- Two types 8 bit & 16 bit
- 8-bit ISA 62 pins and 16-bit ISA had a 98 pins



AGP Slot

Accelerated Graphics Port / Advanced Graphics Port



Has 124 pins

Northbridge Chipset

- also known as the **memory controller hub**
- typically handles communications between the [CPU](#), [RAM](#), [AGP](#) or [PCI Express](#)



Southbridge Chipset

- also known as the **I/O Controller Hub (ICH)**
- is the chip that controls all of the computers I/O functions, such as [USB](#), audio, serial, the system [BIOS](#), the [ISA bus](#), and the [IDE](#) channels.



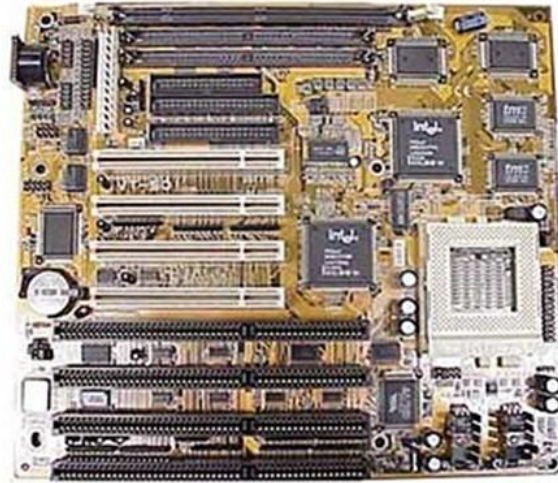
FORM FACTOR

FORM FACTOR

- The form factor of a motherboard determines the specifications for its general shape and size.
- It also specifies what type of case and power supply will be supported, the placement of mounting holes, and the physical layout and organization of the board.
- Form factor is especially important if you build your own computer systems and need to ensure that you purchase the correct case and components

Baby AT (BAT) Form Factor

- The Baby AT was the standard in the PC industry from roughly 1993-1997.
- a smaller version of the AT, only 9" wide and 13" deep, with newer, smaller components. It was a more compact board, but had the same drawbacks as the AT.



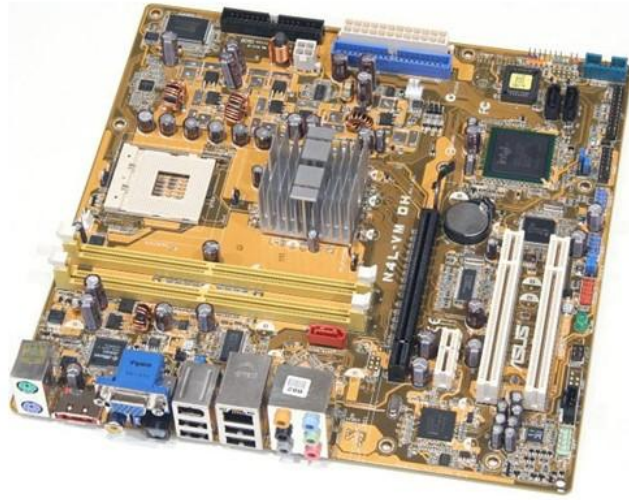
ATX Form Factor (Advanced Technology Extended)



- was designed to overcome the problems associated with the AT/Baby AT form factors.
- the processor and memory are arranged at a right angle to the expansion slots, allowing room for the use of full-length expansion cards.
- 12 inch wide by 9.6 inch long

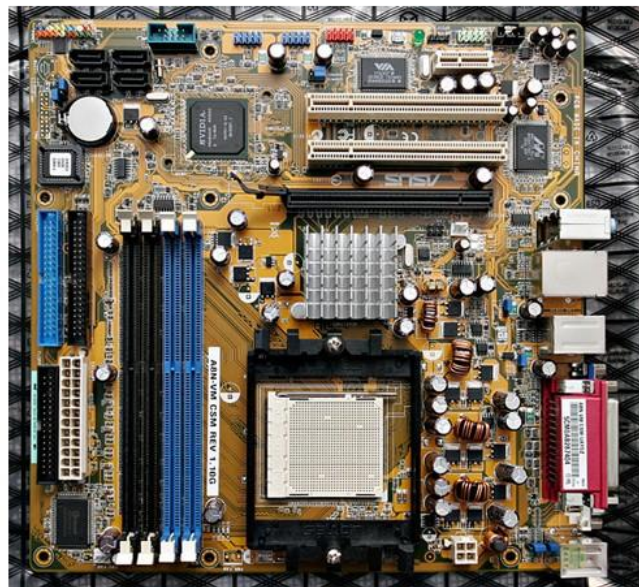
Mini ATX

- Smaller version of a full sized ATX board
- Both designs, parallel, serial, PS/2 keyboard and mouse ports are located on a double-height I/O shield near the rear
- 11.2 inches wide by 8.2 inches long



Micro ATX

- Smaller version of ATX intended for compact low cost consumer systems with limited expansion potential
- Size of the board is 9.6 inch square



LPX Form Factor

- Low Profile eXtension
- 8.67" x 9.25"
- Originally developed by Western Digital
- Expansion Slots are located on a central riser card to be mounted horizontally



NLX Form factor

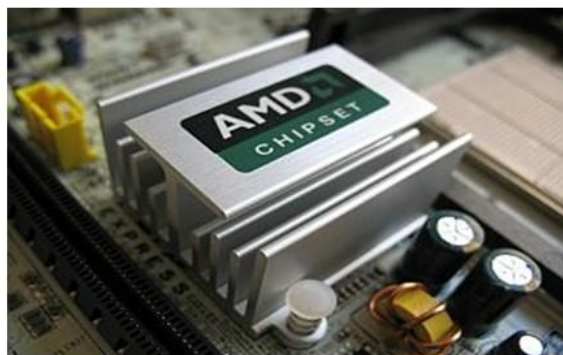
- New Low Profile Extended
- Is a form factor from Intel for PC motherboards
- Introduced in 1997
- 6.6 inch wide by 13 inch long
- All expansion slots, power cables and peripheral are located on an edge-mounted riser card.



CHIPSET

CHIPSET

- is a group of integrated circuits, or chips, that are designed to work together, and are usually marketed as a single product.
- *Northbridge* and the *Southbridge*



Northbridge Chipset

- also known as the **memory controller hub**
- typically handles communications between the CPU, RAM, AGP or PCI Express

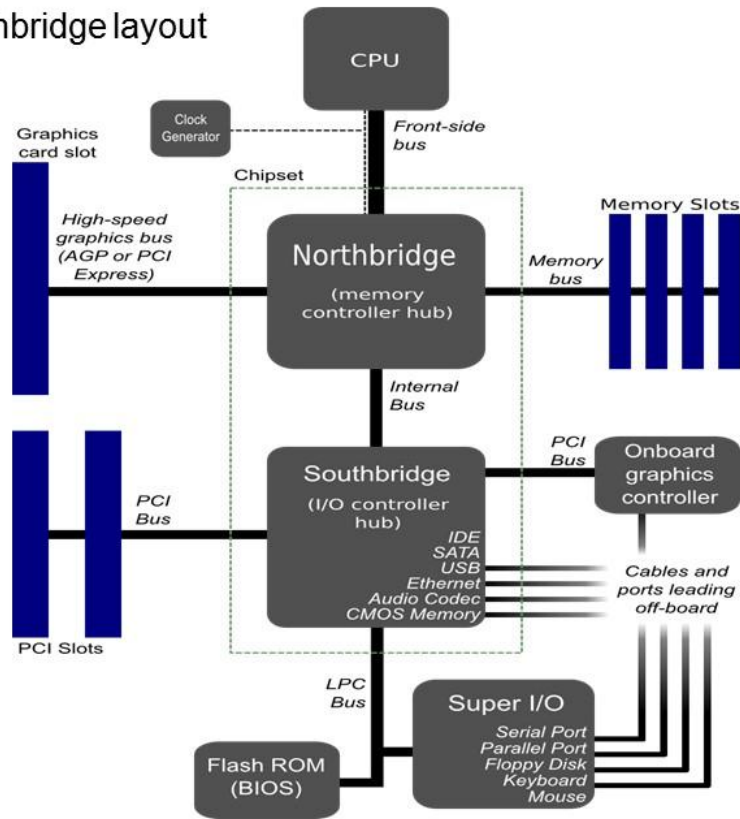


Southbridge Chipset

- also known as the I/O Controller Hub (ICH)
- is the chip that controls all of the computers I/O functions, such as USB, audio, serial, the system BIOS, the ISA bus, and the IDE channels.



A typical north/southbridge layout



Types of Motherboard Buses

- **Processor Bus** - connection between the processor and its primary cache
- **System Bus** – the primary pathway between the CPU, memory and the high speed peripherals to which expansion buses such as ISA, EISA, PCI and VL bus (**VESA Local-BUS**) **Video Electronics Standard Association** can connect

DUAL Independent Bus Architecture

- **Front Side Bus** - is the data transfer bus that carries information between the CPU and the northbridge of the Motherboard
- **Back side bus** - is a computer bus used to connect the CPU to CPU cache memory
- **Local Bus** - is a computer bus that connects directly, or almost directly, from the CPU to one or more slots on the expansion bus.
- **Expansion Bus** - is made up of electronic pathways which move information from your CPU and RAM to all of your other peripheral devices such as a microphone, monitor, telephone line, and printer, which all connect to ports on the back of your computer.

TYPES OF EXPANSION BUSES

- ISA
- MCA (Micro Channel Architecture)
- EISA (Extended Industry Standard Architecture)
- VESA
- PCI (Peripherals Component Inter Connect)
- PCI – Express
- AGP (Accelerated Graphics Port)
- USB (Universal Serial Bus)
- PCMCIA (Personal Computer Card International Association)
- CMOS Battery
- Keyboard Controller/Adapter

ISA

- **Industry Standard Architecture**
- Two types 8 bit & 16 bit
- **8-bit ISA** 62 pins and 16-bit ISA had a 98 pins
- Transfer rate is up to 16MBps
- Released in 1981 by [IBM](#), dominant in the marketplace in the 1980s.



MCA

- **Micro Channel Architecture**
- was a proprietary 16- or 32-bit parallel computer bus created by IBM in the 1980s their new PS/2 computers.
- Does not support ISA cards
- Transfer rate is at 20 MBps



32 Bit



16 Bit

EISA

- **Extended Industry Standard Architecture**
- introduced and developed by the "Gang of Nine" in 1984
- Allow ISA card to be installed
- Transfer rate of 20MBps
- Has 98 + 100 inlay pins



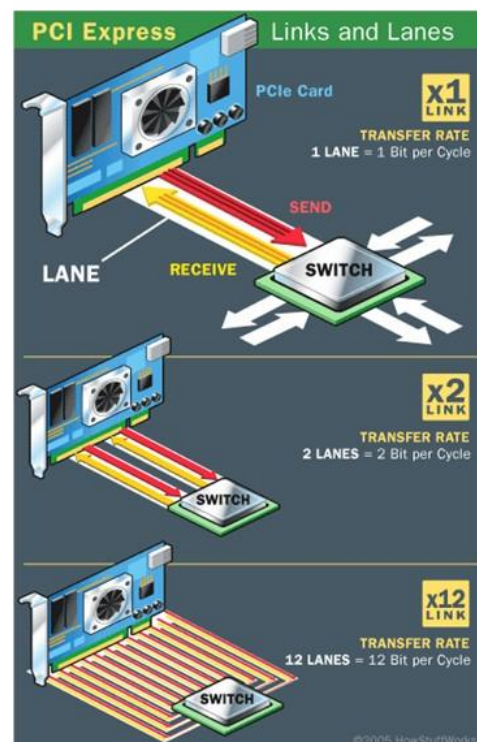
VESA Local Bus

- Video Electronics Standards Association
- was mostly used in personal computers. VESA Local Bus worked alongside the ISA bus
- It has 112 pins
- was designed as a stopgap solution to the problem of the ISA bus's limited bandwidth



PCI EXPRESS

- is a [computer expansion card](#) interface format introduced by [Intel](#) in 2004.
- Each lane of a PCI Express connection contains two pairs of wires -- one to send and one to receive.
- A **x1 connection**, the smallest PCIe connection, has one lane made up of four wires. It carries one bit per cycle in each direction. A x2 link contains eight wires and transmits two bits at once, a x4 link transmits four bits, and so on. Other configurations are x12, x16 and x32.



PCI

- Peripheral Component Interconnect
- a computer bus for attaching peripheral devices to a computer motherboard
- used in PCs include: network cards, sound cards, modems, extra ports such as USB or serial, TV tuner cards and disk controllers.
- 32 **bits** and can handle only 5 devices at a time.



USB

- Universal Serial Bus
- was designed to allow many peripherals to be connected using a single standardized interface socket and to improve the plug-and-play capabilities by allowing devices to be connected and disconnected without rebooting the computer
- USB can connect computer peripherals such as mice, keyboards, PDAs, gamepads and joysticks, scanners, digital cameras, printers, personal media players, and flash drives.



USB Version

Prereleases

- **USB 0.7:** Released in November 1994.
- **USB 0.8:** Released in December 1994.
- **USB 0.9:** Released in April 1995.
- **USB 0.99:** Released in August 1995.
- **USB 1.0**
- **USB 1.0:** Released in January 1996.
Specified data rates of **1.5 Mbit/s (Low-Speed)** and **12 Mbit/s (Full-Speed)**.
- **USB 1.1:** Released in September 1998.
- **USB 2.0:** Released in April 2000.
Added higher maximum speed of **480 Mbit/s** (now called **Hi-Speed**).

PCMCIA

- **Personal Computer Memory Card International Association**
- is an international [standards body](#) that defines and promotes the [PC Card](#) (formerly known as "PCMCIA card") and [ExpressCard](#) standards.
- These cards can be used for wireless connectivity, modem and other functions in laptop/notebook



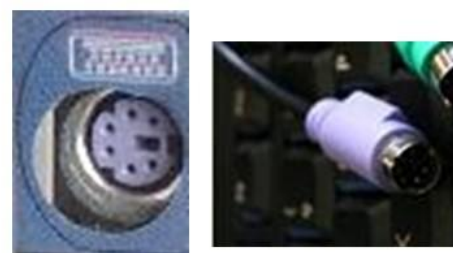
CMOS Battery

- Complementary metal–oxide–semiconductor
- refers to the [memory](#) on a [personal computer motherboard](#) containing [BIOS](#) settings and sometimes the code used to initialize the computer and load the operating system.
- The memory and real-time clock are generally powered by a [CR2032 lithium coin cell](#).
- These cells last two to ten years, depending on the type of motherboard

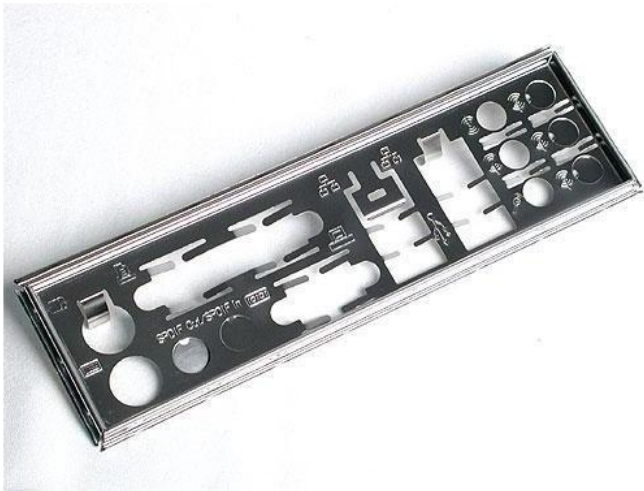


Keyboard Controller Adapter

- 8042 keyboard controller which resides on the system board
- It has 5 pin din and 6 pin din
- DIN (Deutsche Industrial Normale) - is a series of uniformity standards developed in Germany, which apply to commonly manufactured items



Rear I/O Panel



Types Of Screws

